Protocol v3.1 — statistical appendix

Proofs of the sensitivity, concentration and inference claims (S0–S6)

Technical appendix to `protocol_v31.tex`

Abstract

We prove the statistical claims of Protocol v3.1. Proposition S1 gives the exact replace-one ℓ_2 sensitivity of the released vector $(\mathrm{triu}\,\mathbf{B}, \mathbf{C}, \mathbf{c}_R, \mathbf{c}_O, \mathbf{y}^\top\mathbf{y})$ in all three regimes, including a proof that the norms sit at their bounds (and where they do not), the triu-versus-Frobenius question, the case $d_O = 1$, the sensitivities $\Delta_{\neg yy}$, Δ_G, Δ_y, and the set of bounds on which $\mathbf{y}^\top\mathbf{y}$ is free. Lemma S2 bounds the operator norm of the masked symmetric noise matrix; Lemma S3 analyses the two-stage choice of the ridge; Lemma S4 bounds $\|\boldsymbol{\beta}\|$; Proposition S5 states the coverage of both intervals of Algorithm 6, separating exact, first-order and asymptotic parts; Lemma S6 analyses the parametric-bootstrap RSS correction. Sections S7–S10, added after a second reading of the companion book on the protocol, prove the exact stage-1 radius and its transfer to the lattice noise, a bracket for the switch-off of the ridge and a bound on the downward bias of $\lambda_{\min}$, the mechanism bias with explicit constants, the bias of the plug-in residual variance used in the FWS comparison, a valid bias-aware interval under binding clipping, and a closed form for the full-record sensitivity. Section 14 lists every place where the specification is false, imprecise or needs an extra condition, with corrected statements. Every step that is not fully rigorous is marked [first order], [asymptotic], [numerical] or [open].

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1 Summary

2 Setting, notation and the noise model

Notation. We follow `protocol_v31.tex`. R's block has $d_A := d_R + 1$ coordinates (intercept included) and O's block d_O, so $p = d_A + d_O$. For a matched record, $\mathbf{a} := \dot{\mathbf{x}} \in \mathbb{R}^{d_A}$ is R's full row ($\|\mathbf{a}\| \le B_R$, $a_1 = 1$), $\mathbf{x} := \mathbf{x}_O \in \mathbb{R}^{d_O}$ ($\|\mathbf{x}\| \le B_O$) and y ($|y| \le B_y$). We write

$$r = B_R^2, \quad o = B_O^2, \quad w = B_y^2, \quad b = B_y.$$

Π_O is the coordinate projection on O's block. The mask $\chi \in \{0,1\}^{p\times p}$ has $\chi_{ij} = 0$ iff $i, j \le d_A$. For the ridge chosen by Algorithm 6 we write

$$\mathbf{G}_\lambda = \mathbf{G} + \lambda\Psi, \qquad \mathbf{P}_\lambda = \mathbf{G}_\lambda^{-1}, \qquad \boldsymbol{\theta}_\lambda = \mathbf{P}_\lambda \mathbf{c}, \qquad \vartheta_0 = \frac{\lambda_{\min}(\mathbf{G}_\lambda)}{2\sigma_2\sqrt{p}}, \qquad \hat{\vartheta} = \frac{\lambda_{\min}(\hat{\mathbf{G}}_2 + \lambda\Psi)}{2\sigma_2\sqrt{p}}.$$

ϑ_0 is the (unobservable) validity ratio of the ridge at the true Gram matrix; $\hat{\vartheta}$ is the gate statistic of Algorithm 6, line 7. The notation table of the specification calls $\hat{\vartheta}$ "the validity ratio"; the guarantee of Lemma S3 is about ϑ_0. σ_{yy} is the noise s.d. of $\widehat{yy}$ (σ_2 in joint mode, σ_y in separate mode). $z_a = \Phi^{-1}(a)$, ϕ and Φ are the standard normal density and distribution function, and $Q = 1 - \Phi$.

The masked noise model. A *masked symmetric noise matrix* with scale σ is $\mathbf{E} = \mathbf{E}^\top \in \mathbb{R}^{p\times p}$ with $E_{ij} = 0$ when $\chi_{ij} = 0$ and $\{E_{ij} : i \le j,\ \chi_{ij} = 1\}$ independent, each $\mathcal{N}(0, \sigma^2)$ (the diagonal of the O block included, since triu $\mathbf{B}$ is released entry by entry). Its covariance is

$$\mathrm{Cov}(E_{ij}, E_{kl}) = \sigma^2\chi_{ij}\big(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \delta_{ij}\delta_{ik}\delta_{il}\big). \tag{1}$$

Stage s releases $\hat{\mathbf{G}}_s = \mathbf{G} + \mathbf{E}_s$ and (stage 2) $\hat{\mathbf{c}}_2 = \mathbf{c} + \mathbf{f}_2$ with $\mathbf{f}_2 \sim \mathcal{N}(0, \sigma_2^2\mathbf{I}_p)$ (every entry of $\mathbf{c}_R$ and $\mathbf{c}_O$ depends on $\mathbf{y}$), and $\widehat{yy} = \mathbf{y}^\top\mathbf{y} + g$ with $g \sim \mathcal{N}(0, \sigma_{yy}^2)$; the noises of different releases are independent and independent of the data. For $\boldsymbol{\theta} \in \mathbb{R}^p$ we use

$$\mathbf{K}(\boldsymbol{\theta}) := \sigma^{-2}\,\mathrm{Cov}(\mathbf{E}\boldsymbol{\theta}), \qquad K(\boldsymbol{\theta})_{ik} = \delta_{ik}\Big(\sum_j \chi_{ij}\theta_j^2 - \chi_{ii}\theta_i^2\Big) + \chi_{ik}\theta_i\theta_k, \tag{2}$$

Item	What is proved	Status
S0	The slot noise is the image of the lattice discrete Gaussian under a scaled orthogonal map: exactly uncorrelated, σ_s^2-subgaussian, and within total variation $N(0.0202\,\sigma_c^{-2} + 0.0315\,\sigma_c^{-3} + e^{-2\pi^2\sigma_c^2})$ of i.i.d. $\mathcal{N}(0,\sigma_s^2)$ plus a perturbation of norm $\le N/(2\sqrt{2}\,\Omega)$.	exact (TV constant now proved)
S1	Exact Δ in regimes 1–2 (closed form), exact finite algorithm in regime 3; norms at bounds; triu = Frobenius supremum for $d_O \ge 2$; $d_O = 1$ exact reduction and closed-form Δ_G; $\Delta_{\neg yy}$ in closed form in all regimes; $\mathbf{y}^\top\mathbf{y}$ free if $w \le 2(r+o)$, costly if $w > 3(r+o)$.	exact; boundary of the free set [numerical]
S2	$\mathbb{E}\lambda_{\max}(\mathbf{E}) \le 2\sigma\sqrt{p}$; $\sqrt{2}\sigma$-Lipschitz concentration; $\Pr[\lambda_{\max}(\mathbf{E}) \ge r_1/\sigma_1 \cdot \sigma] \le \alpha_L$.	exact (one-sided); $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} \le 2\sigma\sqrt{p}$ [open]
S3	Measurability and independence; on Ω_L: $\lambda_{\min}(\mathbf{G}+\lambda\Psi) \ge 2\vartheta^\star\sigma_2\sqrt{p}$, $\hat{L} \le \lambda_{\min}(\mathbf{G})$; closed-form λ_O; gate-failure bound.	exact
S4	$\|\boldsymbol{\beta}\|^2 \le \mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}]/\lambda_{\min}(\mathbf{G})$, also for the projection estimand; U_{yy} valid at level $2\alpha_U$ as written, α_U after correction.	exact (needs conditional independence)
S5	Exact Gaussian first-order term, exact resolvent remainder identity, rigorous finite-sample coverage inequality for the oracle Wald interval; AKK inequality and union bound for the β_j interval.	plug-in [first order]; normality of sampling term [asymptotic]
S6	Second-order RSS bias in closed form; Corr removes it up to a relative $O(\vartheta_0^{-2})$ remainder; bootstrap covariance consistent to relative $O_p(\vartheta_0^{-1})$.	[first order]; constants not explicit
S7	Exact stage-1 radius: exact level $(M-j)/(M+1)$ with fresh Monte Carlo draws, confidence c over the draws; transfer to the protocol with the proved TV constant and offset bound.	exact
S8	Bracket for the switch-off, tight among Gram matrices with a given $\lambda_{\min}$; spectral lower bound on $\mathbb{E}\lambda_{\min}(\mathbf{G}+\mathbf{E})$ by Sudakov–Fernique.	exact; a spectrum-sharp tail bound [open]
S9	Leading-order mechanism bias, ridge version and decay in λ, with explicit remainder constants.	exact remainder bounds
S10	Bias of FWS's plug-in RSS with explicit remainder; a valid bias-aware interval under binding clipping with a public variance bound; closed form of the full-record sensitivity for blocks of dimension ≥ 2.	exact; data-driven variance under clipping [open]

Table 1: What is proved. Corrections to the specification are collected in Section 14; open steps in Section 15.

and $\mathbf{M}_\chi(\mathbf{P}) := \sigma^{-2}\mathbb{E}[\mathbf{EPE}] = \chi \circ \mathbf{P} + \mathrm{diag}_i\big(\sum_j \chi_{ij}P_{jj} - \chi_{ii}P_{ii}\big)$, both direct consequences of (1). $0 \preceq \mathbf{K}(\boldsymbol{\theta})$ and $\|\mathbf{K}(\boldsymbol{\theta})\| \le 2\|\boldsymbol{\theta}\|^2$.

Assumption 2.1 (Gaussian slot model, N)**.** Every released entry equals its exact value plus an independent $\mathcal{N}(0,\sigma_s^2)$ variable, independently across releases and of the data.

All statistical statements below are proved under Assumption 2.1. Lemma S0 shows how they transfer to the actual mechanism, whose noise is the linear image of a lattice discrete Gaussian.

Lemma 2.2 (S0: the slot noise)**.** *Let $\eta \sim \mathcal{N}_{\mathbb{Z}}(0,\sigma_c^2)^N$ and let $\boldsymbol{\xi}^{\mathrm{raw}} \in \mathbb{R}^m$ be the noise in m released slot values (real parts of distinct slots, or any m distinct coordinates of $(\mathrm{Re}\,\tau(\eta), \mathrm{Im}\,\tau(\eta))/\Omega$). Write $\sigma_s = \sigma_c\sqrt{N/2}/\Omega$.*

(a) $\boldsymbol{\xi}^{\mathrm{raw}} = (\sigma_s/\sigma_c)\,\mathbf{U}\eta$ for a matrix $\mathbf{U} \in \mathbb{R}^{m\times N}$ with orthonormal rows. Hence the coordinates of $\boldsymbol{\xi}^{\mathrm{raw}}$ are exactly *uncorrelated, with common variance $\sigma_s^2\,v(\sigma_c)/\sigma_c^2$, where $v(\sigma) = \mathrm{Var}\,\mathcal{N}_{\mathbb{Z}}(0,\sigma^2)$ satisfies $\sigma^2\big(1 - O(\sigma^2 e^{-2\pi^2\sigma^2})\big) \le v(\sigma) \le \sigma^2$; and $\boldsymbol{\xi}^{\mathrm{raw}}$ is σ_s^2-subgaussian: $\mathbb{E}e^{\langle \mathbf{u}, \boldsymbol{\xi}^{\mathrm{raw}}\rangle} \le e^{\sigma_s^2\|\mathbf{u}\|^2/2}$ for all $\mathbf{u}$.*

(b) There is a coupling of η with $\mathbf{Z} \sim \mathcal{N}(0, \sigma_c^2 \mathbf{I}_N)$ such that $\Pr[\eta \neq \lfloor \mathbf{Z} \rceil] \leq N \,\mathrm{TV}_1(\sigma_c)$, where $\mathrm{TV}_1(\sigma) = d_{\mathrm{TV}}\big(\mathcal{N}_{\mathbb{Z}}(0, \sigma^2), \lfloor \mathcal{N}(0, \sigma^2) \rceil\big)$ satisfies, for every $\sigma > 0$,

$$\mathrm{TV}_1(\sigma) \leq \frac{\phi(1)}{12\,\sigma^2} + \frac{V}{48\,\sigma^3} + \sum_{j \geq 1} e^{-2\pi^2\sigma^2 j^2}, \qquad V = 2\phi(0) + 8\phi(\sqrt{3}) = 1.5100,$$

that is, $\mathrm{TV}_1(\sigma) \leq 0.020165\,\sigma^{-2} + 0.031459\,\sigma^{-3} + 1.0001\,e^{-2\pi^2\sigma^2}$ for $\sigma \geq 1$. On that event $\boldsymbol{\xi}^{\mathrm{raw}} = \boldsymbol{\xi}^G + \boldsymbol{\nu}$ with $\boldsymbol{\xi}^G = (\sigma_s/\sigma_c)\mathbf{U}\mathbf{Z} \sim \mathcal{N}(0, \sigma_s^2 \mathbf{I}_m)$ exactly and $\|\boldsymbol{\nu}\| \leq N/(2\sqrt{2}\,\Omega)$.

(c) The decrypted value also carries the flooded error $e_{\mathrm{tot},s}$ and the encoding rounding. On $\{\|e_{\mathrm{tot},s}\| \leq B_{\mathrm{tot},s}\}$ (probability $\geq 1 - \delta_B$) their slot-domain contribution has ℓ_2 norm at most $N/(2\sqrt{2}\,\Omega) + \sqrt{2\pi}\,2^{-\kappa}\sigma_s$, using $\sigma_c \geq 2^{\kappa} B_{\mathrm{tot}}/\sqrt{2\pi}$.

Consequently every statement proved under Assumption 2.1 holds for Protocol v3.1 after adding $\delta_B + N\,\mathrm{TV}_1(\sigma_{c,s})$ (and the wraparound probability excluded by Calibrate*) per release to the failure probabilities, and enlarging every radius, in operator or Frobenius norm, by $\sqrt{2}\,\bar{\nu}$ with $\bar{\nu} := N/(\sqrt{2}\,\Omega) + \sqrt{2\pi}\,2^{-\kappa}\sigma_s$ (the factor $\sqrt{2}$ accounts for the mirrored off-diagonal entries of $\hat{\mathbf{B}}$). With the indicative parameters ($N = 2^{14}$, $\log_2 \Omega \approx 52.5$, $\kappa = 40$, $\sigma_c \approx 2^{50}$) both additions are below 10^{-11} in relative terms.*

Proof. (a) For $m(X) = \sum_k \eta_k X^k$, slot j is $\tau(m)_j = \sum_k \eta_k \zeta^{e_j k}$ with $\zeta = e^{i\pi/N}$ and $e_j = 5^j \bmod 2N$ odd. For $j \neq l$, $\sum_{k=0}^{N-1} \cos(\pi k e_j/N)\cos(\pi k e_l/N) = \frac{1}{2}\sum_k [\cos(\pi k(e_j - e_l)/N) + \cos(\pi k(e_j + e_l)/N)] = 0$, because $e_j \pm e_l$ is even and non-zero modulo $2N$ (the subgroup generated by 5 does not contain -1), and $\sum_{k<N} e^{i\pi k m/N} = (1 - e^{i\pi m})/(1 - e^{i\pi m/N}) = 0$ for even $m \not\equiv 0$. The same computation gives $\sum_k \cos^2 = \sum_k \sin^2 = N/2$ and $\sum_k \cos \cdot \sin = 0$. Hence the N real numbers $(\mathrm{Re}\,\tau(m), \mathrm{Im}\,\tau(m))$ equal $\sqrt{N/2}\,\mathbf{U}_0\eta$ for an orthogonal $\mathbf{U}_0$ (this is the identity $\|\tau^{-1}(\mathbf{z})\| = \sqrt{2/N}\|\mathbf{z}\|$ of Lemma 1 of the specification), and $\mathbf{U}$ is a set of m rows of $\mathbf{U}_0$. The variance and subgaussian bounds for $\mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$ are in Canonne, Kamath and Steinke [6] (Section 2); the lower bound on v follows from Poisson summation, $v(\sigma) = \sigma^2 \frac{1 + \sum_{m \neq 0}(1 - 4\pi^2\sigma^2 m^2) e^{-2\pi^2\sigma^2 m^2}}{1 + \sum_{m \neq 0} e^{-2\pi^2\sigma^2 m^2}}$. Subgaussianity of $\boldsymbol{\xi}^{\mathrm{raw}}$ follows because $\langle \mathbf{u}, \boldsymbol{\xi}^{\mathrm{raw}} \rangle = (\sigma_s/\sigma_c)\langle \mathbf{U}^{\top}\mathbf{u}, \eta \rangle$ with $\|\mathbf{U}^{\top}\mathbf{u}\| = \|\mathbf{u}\|$ and independent coordinates.

(b) Couple coordinatewise with a maximal coupling; a union bound gives $N\,\mathrm{TV}_1$. Write $p_k \propto e^{-k^2/2\sigma^2}$ and $q_k = \int_{k-1/2}^{k+1/2} \phi_\sigma$, so $\mathrm{TV}_1 = \frac{1}{2}\sum_k |p_k - q_k|$. First, $p_k = \phi_\sigma(k)/\sum_l \phi_\sigma(l)$ and, by Poisson summation, $\sum_l \phi_\sigma(l) = 1 + 2\epsilon$ with $\epsilon = \sum_{j \geq 1} e^{-2\pi^2\sigma^2 j^2}$; so $\sum_k |p_k - \phi_\sigma(k)| = |1 - \sum_l \phi_\sigma(l)| = 2\epsilon$. Second, $q_k - \phi_\sigma(k) = \int_{-1/2}^{1/2} (\phi_\sigma(k + u) - \phi_\sigma(k) - u\phi'_\sigma(k))\,du$ (the linear term integrates to zero), so $|q_k - \phi_\sigma(k)| \leq \frac{1}{24}\sup_{I_k} |\phi''_\sigma|$ with $I_k = [k - \frac{1}{2}, k + \frac{1}{2}]$. For any function f of bounded variation, $\sup_{I_k} |f| \leq \int_{I_k} |f| + \mathrm{TV}_{I_k}(|f|)$, so $\sum_k \sup_{I_k} |\phi''_\sigma| \leq \int |\phi''_\sigma| + \mathrm{TV}(|\phi''_\sigma|) = 4\phi(1)\sigma^{-2} + V\sigma^{-3}$: $|\phi''(x)| = \phi(x)|x^2 - 1|$ for the standard density has zeros at ± 1 and local maxima $\phi(0)$ at 0 and $2\phi(\sqrt{3})$ at $\pm\sqrt{3}$, so its total variation is $2\phi(0) + 8\phi(\sqrt{3})$, and $\int |\phi''| = 4\phi(1)$. Halving the sum of the two parts gives the bound. It is within a factor 1.02 of the exact value at $\sigma = 100$ and within a factor 3.2 at $\sigma = 1$ (exact high-precision evaluation, `experiments/proofs_check/tv1_cuckoo.py`); at the evaluation setting ($N = 2^{14}$, $\sigma_c \approx 2^{65.3}$) $N\,\mathrm{TV}_1(\sigma_c) \leq 2^{-122.2}$. (An earlier version gave only the leading constant, with the $O(\sigma^{-3})$ term checked numerically; so does the companion book's Lemma `rel.slot`.) Finally $\|\lfloor \mathbf{Z} \rceil - \mathbf{Z}\| \leq \sqrt{N}/2$ and $\|(\sigma_s/\sigma_c)\mathbf{U}\|_{\mathrm{op}} = \sqrt{N/2}/\Omega$.

(c) The rounding of `Ecd` has sup-norm $\leq 1/2$; the slot image of a ring element e has norm $\sqrt{N/2}\|e\|/\Omega$, and $\sqrt{N/2}B_{\mathrm{tot}}/\Omega \leq \sqrt{N/2}\sqrt{2\pi}\sigma_c 2^{-\kappa}/\Omega = \sqrt{2\pi}2^{-\kappa}\sigma_s$. □

3 Proposition S1: replace-one sensitivity

3.1 The change vector

Replacing $(\mathbf{x}, y)$ by $(\mathbf{x}', y')$ in a matched record with R-row $\mathbf{a}$ changes $(\operatorname{triu}\mathbf{B}, \mathbf{C}, \mathbf{c}_R, \mathbf{c}_O, \mathbf{y}^\top\mathbf{y})$ by

$$\big(\operatorname{triu}\mathbf{D},\ \mathbf{a}(\mathbf{x}-\mathbf{x}')^\top,\ \mathbf{a}(y-y'),\ \mathbf{x}y-\mathbf{x}'y',\ y^2-y'^2\big), \qquad \mathbf{D} := \mathbf{x}\mathbf{x}^\top - \mathbf{x}'\mathbf{x}'^\top.$$

A change in an unmatched (or dummy) row changes nothing, because its O-side entries are multiplied by $b_t = 0$ or by $\dot{\mathbf{x}}_t = \mathbf{0}$ (Algorithm 4). Hence $\Delta^2 = \max_{\mathcal{D}} \Phi_{\text{triu}}$ with

$$\Phi_{\text{triu}}(\mathbf{a}, \mathbf{x}, \mathbf{x}', y, y') = \|\operatorname{triu}\mathbf{D}\|^2 + \|\mathbf{a}\|^2\big(\|\mathbf{x}-\mathbf{x}'\|^2 + (y-y')^2\big) + \|\mathbf{x}y - \mathbf{x}'y'\|^2 + (y^2-y'^2)^2,$$
$$\mathcal{D} = \{\|\mathbf{a}\| \le B_R,\ \|\mathbf{x}\|, \|\mathbf{x}'\| \le B_O,\ |y|, |y'| \le B_y\}$$

(compact, so the supremum is a maximum). Φ_F denotes the same expression with $\|\operatorname{triu}\mathbf{D}\|^2$ replaced by $\|\mathbf{D}\|_F^2$, and $\Phi_{\text{triu}}^{\neg yy}$, $\Phi_F^{\neg yy}$ the expressions without the last term. The intercept constraint $a_1 = 1$ is irrelevant: the maximum has $\|\mathbf{a}\| = B_R \ge 1$, which is attained by $\mathbf{a} = (1, \mathbf{z})$ with $\|\mathbf{z}\|^2 = r - 1$ whenever $d_R \ge 1$ (if $d_R = 0$ then $B_R = 1$). Δ is the supremum over all cohorts; for a given cohort the sensitivity can only be smaller.

Define on $[-o, o] \times [-b, b]$

$$F(s,t) = 2o^2 - 2s^2 + 2r(o-s) + r(b-t)^2 + o(w+t^2) - 2bts + (w-t^2)^2, \tag{3}$$
$$F^{\neg yy}(s,t) = F(s,t) - (w-t^2)^2, \qquad s^*(t) = \operatorname{clip}\big(-(r+bt)/2;\ -o,\ o\big), \qquad G(t) = F(s^*(t), t).$$

F is the function of the previous paper (with $s = oc$), so the reduction below is the missing justification of its use.

3.2 Statement

Theorem 3.1 (S1)**.** *Let $d_O \ge 2$ and $b > 0$ ($b = 0$ is covered by setting $w = 0$).*

(i) ***Reduction.*** $\Delta^2 = \max_{|s|\le o, |t|\le b} F(s,t) = \max_{|t|\le b} G(t)$.

(ii) ***Regime 1****, $|r - w| \le 2o$: $\Delta^2 = \frac{1}{2}(2o + r + w)^2 + 2rw$, attained exactly at $t = -b$, $s = (w - r)/2$.*

(iii) ***Regime 2****, $r - w \ge 2o$: $\Delta^2 = 4r(o + w)$, attained exactly at $t = -b$, $s = -o$.*

(iv) ***Regime 3****, $w - r > 2o$: with $t_1 = -(r + 2o)/b$ and $t_2 = (2o - r)/b$ one has $-b < t_1 < t_2 < b$ and G is C^1 and piecewise quartic:*

$$G_A(t) = t^4 + (r + o - 2w)t^2 - 2b(r + o)t + (r + o)w + w^2 \qquad t \in [-b, t_1]\ (s^* = o),$$
$$G_B(t) = t^4 + (r + o - \tfrac{3}{2}w)t^2 - rbt + 2o^2 + 2ro + (r + o)w + w^2 + \tfrac{1}{2}r^2 \qquad t \in [t_1, t_2]\ (|s^*| < o),$$
$$G_C(t) = t^4 + (r + o - 2w)t^2 + 2b(o - r)t + 4ro + (r + o)w + w^2 \qquad t \in [t_2, b]\ (s^* = -o).$$

$\Delta^2 = \max_{t\in\mathcal{T}} G(t)$ for the finite set $\mathcal{T} = \{-b, t_1, t_2, b\} \cup \{$real roots of G_A', G_B', G_C' in their intervals$\}$; this is exact (three cubics). $t = -b$ is a strict local maximizer with $G(-b) = 4w(r + o)$; if $\Delta^2 > 4w(r + o)$, every maximizer has $|t^*| < b$.

(v) ***Norms.*** *At every maximizer $\|\mathbf{a}\| = B_R$, $\|\mathbf{x}\| = \|\mathbf{x}'\| = B_O$ and $\max(|y|, |y'|) = B_y$. Both $|y| = |y'| = B_y$ in regimes 1–2; in regime 3 this fails whenever $\mathbf{y}^\top\mathbf{y}$ is not free ($|y'| = |t^*| < B_y$).*

(vi) triu ***versus Frobenius.*** $\max_{\mathcal{D}} \Phi_{\text{triu}} = \max_{\mathcal{D}} \Phi_F$ *(and the same without $\mathbf{y}^\top\mathbf{y}$); a configuration attains the* triu *maximum iff it attains the Frobenius maximum and $\mathbf{D}$ is diagonal.*

(vii) ***Worst case.*** *Let $s^\dagger$, $t^\dagger$ be a maximizer of F and $c = s^\dagger/o$, $\alpha_\pm = \frac{1}{2}(\sqrt{1+c} \pm \sqrt{1-c})$. A worst-case pair is $\|\mathbf{a}\| = B_R$, $y = B_y$, $y' = t^\dagger$,*

$$\mathbf{x} = B_O(\alpha_+\mathbf{e}_1 + \alpha_-\mathbf{e}_2), \qquad \mathbf{x}' = B_O(\alpha_-\mathbf{e}_1 + \alpha_+\mathbf{e}_2),$$

for which $\mathbf{D} = o\sqrt{1-c^2}\,(\mathbf{e}_1\mathbf{e}_1^\top - \mathbf{e}_2\mathbf{e}_2^\top)$. *In regime 1:* $y' = -y$, $\cos\angle(\mathbf{x},\mathbf{x}') = -(r-w)/(2o)$; *in regime 2:* $y' = -y$, $\mathbf{x}' = -\mathbf{x}$ *(*$\mathbf{D} = 0$*).*

(viii) ***Other sensitivities.*** $\Delta^2_{\neg yy} = \frac{1}{2}(2o + r + w)^2 + 2rw$ *if* $|r - w| \le 2o$ *and* $\Delta^2_{\neg yy} = 4M(o + m)$ *otherwise, with* $M = \max(r, w)$ *and* $m = \min(r, w)$. $\Delta_G = \Delta(B_R, B_O, 0)$ *equals* $(2o + r)/\sqrt{2}$ *if* $r \le 2o$ *and* $2\sqrt{ro}$ *otherwise.* $\Delta_y = w$.

(ix) ***When*** $\mathbf{y}^\top\mathbf{y}$ ***is free.*** $\Delta = \Delta_{\neg yy}$ *iff* $\max_{|t|\le b} G(t) \le 4w(r + o)$ *or* $w \le r + 2o$. *In particular it is free whenever* $w \le 2(r + o)$ *(this covers regimes 1, 2 and part of regime 3), and costly whenever* $w > 3(r + o)$. *For* $o = 0$ *the exact threshold is* $w \le \frac{11+5\sqrt{5}}{8}\, r \approx 2.7725\, r$.

The specification's DeltaRep ("exact for $d_O \ge 2$") is therefore correct in regimes 1–2 and, after the reduction (i), in regime 3, where (iv) replaces the grid search by an exact finite computation. For $d_O = 1$ see Proposition 3.5.

3.3 Two lemmas

Lemma 3.2 (triu versus Frobenius)**.** *For all* $\mathbf{x}, \mathbf{x}'$: $\|\operatorname{triu}\mathbf{D}\|^2 = \frac{1}{2}\big(\|\mathbf{D}\|_F^2 + \sum_j D_{jj}^2\big) \le \|\mathbf{D}\|_F^2$, *with equality iff* $\mathbf{D}$ *is diagonal. If* $d_O \ge 2$, *there is an orthogonal* $\mathbf{Q}$ *such that* $\mathbf{Q}\mathbf{D}\mathbf{Q}^\top$ *is diagonal; replacing* $(\mathbf{x}, \mathbf{x}')$ *by* $(\mathbf{Q}\mathbf{x}, \mathbf{Q}\mathbf{x}')$ *changes* $\mathbf{D}$ *into* $\mathbf{Q}\mathbf{D}\mathbf{Q}^\top$ *and leaves every other term of* Φ *and the constraints unchanged. Hence* $\max\Phi_{\text{triu}} = \max\Phi_F$. *For* $d_O = 1$, $\operatorname{triu}\mathbf{D} = \mathbf{D}$.

Proof. The identity holds because $\mathbf{D}$ is symmetric. $\mathbf{D}$ has rank ≤ 2 and range in $\operatorname{span}\{\mathbf{x}, \mathbf{x}'\}$, so $\mathbf{D} = \mu_1\mathbf{q}_1\mathbf{q}_1^\top + \mu_2\mathbf{q}_2\mathbf{q}_2^\top$ with orthonormal $\mathbf{q}_1, \mathbf{q}_2$; any orthogonal $\mathbf{Q}$ with $\mathbf{Q}\mathbf{q}_i = \mathbf{e}_i$ diagonalizes $\mathbf{D}$ (this needs $d_O \ge 2$). The other terms depend on $(\mathbf{x}, \mathbf{x}')$ only through $\|\mathbf{x}\|, \|\mathbf{x}'\|, \mathbf{x}^\top\mathbf{x}'$, which are invariant. Thus $\Phi_{\text{triu}}(\cdot, \mathbf{Q}\mathbf{x}, \mathbf{Q}\mathbf{x}', \cdot) = \Phi_F(\cdot, \mathbf{x}, \mathbf{x}', \cdot) \ge \Phi_{\text{triu}}(\cdot, \mathbf{x}, \mathbf{x}', \cdot)$. □

Lemma 3.3 (Reduction)**.** *Let* $d_O \ge 2$. *With* $\rho = \|\mathbf{a}\|^2$, $u = \|\mathbf{x}\|^2$, $v = \|\mathbf{x}'\|^2$, $s = \mathbf{x}^\top\mathbf{x}'$,

$$\Phi_F = \Upsilon := u^2 + v^2 - 2s^2 + \rho\big(u + v - 2s + (y - y')^2\big) + y^2u + y'^2v - 2yy's + (y^2 - y'^2)^2,$$

and the image of $\mathcal{D}$ *is* $\{0 \le \rho \le r,\ 0 \le u, v \le o,\ s^2 \le uv,\ |y|, |y'| \le b\}$.

(a) Υ *is non-decreasing in* ρ, *strictly unless* $(\mathbf{x}, y) = (\mathbf{x}', y')$.

(b) For fixed (ρ, v, s, y, y'), Υ *is strictly increasing in* u *on the feasible set* $\{u \in [0, o] : s^2 \le uv\}$, *which is an interval with right end* o. *Hence every maximizer has* $u = o$, *and likewise* $v = o$.

(c) For fixed (ρ, u, v, s), $\Upsilon = \mathcal{Q}(y, y') + (y^2 - y'^2)^2 + \text{const}$ *with* $\mathcal{Q}$ *positive semidefinite; hence* $\Upsilon(\kappa y, \kappa y') \ge \Upsilon(y, y')$ *for* $\kappa \ge 1$, *and every maximizer has* $\max(|y|, |y'|) = b$.

(d) Υ *is invariant under* $(u, y) \leftrightarrow (v, y')$ *and under* $(y, y') \mapsto (-y, -y')$.

Consequently $\Delta^2 = \max F$ *over* $[-o, o] \times [-b, b]$: *set* $\rho = r$, $u = v = o$, $y = b$, $y' = t$.

Proof. For $d_O \ge 2$, every (u, v, s) with $s^2 \le uv$ is realized by some $(\mathbf{x}, \mathbf{x}')$. (a) The coefficient of ρ is $\|\mathbf{x} - \mathbf{x}'\|^2 + (y - y')^2$. (b) $\partial_u\Upsilon = 2u + \rho + y^2 > 0$ (as $\rho \ge 1$ on $\mathcal{D}$); the feasible set in u is $[s^2/v, o]$ (or $[0, o]$ if $v = 0$), which is upward closed. This is the step the earlier proof sketch asserted without proof: monotonicity holds at fixed inner product s, *not* at fixed cosine (at fixed cosine c, $\partial_u(-2s^2) = -2vc^2$ can dominate). (c) $\mathcal{Q}(y, y') = \rho(y - y')^2 + uy^2 + vy'^2 - 2syy'$ has matrix $\left(\begin{smallmatrix} \rho+u & -(\rho+s) \\ -(\rho+s) & \rho+v \end{smallmatrix}\right)$ with non-negative diagonal and determinant $\rho(u + v - 2s) + uv - s^2 \ge 0$. So $\Upsilon(\kappa y, \kappa y') - \Upsilon(y, y') = (\kappa^2 - 1)\mathcal{Q} + (\kappa^4 - 1)(y^2 - y'^2)^2 \ge 0$. If a maximizer had $\max(|y|, |y'|) < b$, scaling would not increase Υ only if $\mathcal{Q}(y, y') = 0$ and $y^2 = y'^2$. Either $y = y' = 0$, and then moving to $(b, -b)$ increases Υ by $\mathcal{Q}(b, -b) = w(4\rho + u + v + 2s) > 0$; or $\det = 0$, which with $\rho > 0$ forces $u = v = s$, i.e. $\mathbf{x} = \mathbf{x}'$, and then $\mathcal{Q} = (\rho + u)(y - y')^2 = 0$ forces $y = y'$ and $\Upsilon = 0$, not a maximum. (d) is immediate. By (d) we may take $|y| = b$ and then $y = b$. □

3.4 Proof of Theorem 3.1

(i) Lemmas 3.2 and 3.3. For fixed t, F is a strictly concave quadratic in s, $F(s,t) = -2s^2 - 2s(r + bt) + H(t)$ with $H(t) = 2o^2 + 2ro + r(b-t)^2 + o(w+t^2) + (w-t^2)^2$, maximized over $[-o, o]$ at $s^*(t)$.

(ii) Relax $s \in \mathbb{R}$: $G_u(t) := \max_{s\in\mathbb{R}} F = H(t) + \frac{1}{2}(r+bt)^2 = t^4 + (r + o - \frac{3}{2}w)t^2 - rbt + K$, with K the constant term of G_B. Using $w - t^2 = (b-t)(b+t)$ and $w^2 - t^4 = (w-t^2)(w+t^2)$,

$$G_u(-b) - G_u(t) = (b+t)\big[rb + (b-t)\big(r + o - \tfrac{w}{2} + t^2\big)\big]. \tag{4}$$

If $w \le r + 2o$ then $r + o - w/2 \ge r/2 > 0$, so (4) is > 0 for $t > -b$. If also $r - w \le 2o$, the relaxed maximizer at $t = -b$, $s = -(r-w)/2$, is feasible. Hence $\Delta^2 = G_u(-b) = 2o^2 + 2o(r+w) + 4rw + \frac{1}{2}(r-w)^2 = \frac{1}{2}(2o + r + w)^2 + 2rw$, with a unique maximizer (strict concavity in s).

(iii) If $r - w \ge 2o$ then $-(r+bt)/2 \le -(r-w)/2 \le -o$ for all $|t| \le b$, so $s^* \equiv -o$ and $G(t) = F(-o,t) = 4ro + r(b-t)^2 + o(b+t)^2 + (w-t^2)^2$. Then

$$G(-b) - G(t) = (b+t)\big[r(3b-t) - o(b+t) - (b-t)^2(b+t)\big].$$

With $r \ge 2o + w$ and $t = b\tau$, the bracket is at least $o(5b-3t) + b^3\big[(3-\tau) - (1-\tau)^2(1+\tau)\big] = o(5b-3t) + b^3(2 + \tau^2 - \tau^3) \ge 2ob + 2b^3 > 0$. So the maximum is $G(-b) = 4r(o+w)$, attained only at $t = -b$.

(iv) $s^*(t) = o$ iff $r + bt \le -2o$ iff $t \le t_1$, and $s^*(t) = -o$ iff $t \ge t_2$; in regime 3, $t_1 > -b$ and $t_2 < b$. Substituting $s = \pm o$ or $s = -(r+bt)/2$ in (3) gives G_A, G_B, G_C. By Danskin's theorem [8] (unique maximizer $s^*(t)$, continuous in t), $G'(t) = \partial_t F(s^*(t), t)$ is continuous, so a maximizer of G on $[-b, b]$ is an endpoint or a zero of G', i.e. a root of the cubic G'_k of the piece that contains it. Further $G_A(-b) = 4w(r+o)$, $G'_A(-b) = -4b(r+o) < 0$, and $G(b) = G_C(b) = 4o(r+w) < 4w(r+o)$ because $w > r$ in regime 3. So if $\max G > 4w(r+o)$ every maximizer lies in $(-b, b)$.

(v) From Lemma 3.3 and the uniqueness statements in (ii)–(iv).

(vi) Lemma 3.2. If $\mathbf{D}$ is not diagonal at a Frobenius maximizer, the triu value is strictly smaller.

(vii) With $\|\mathbf{x}\| = \|\mathbf{x}'\| = B_O$ and $\mathbf{x}^\top\mathbf{x}' = oc$, $\mathbf{D}$ has eigenvalues $\pm o\sqrt{1-c^2}$ and triu attains $\|\mathbf{D}\|_F$ iff the eigenvectors are coordinate vectors; the stated pair has $\alpha_+^2 + \alpha_-^2 = 1$, $2\alpha_+\alpha_- = c$ and $\alpha_+^2 - \alpha_-^2 = \sqrt{1-c^2}$, which gives the displayed diagonal $\mathbf{D}$. A pair with the right norms and angle but non-diagonal $\mathbf{D}$ is *not* worst case: for $\mathbf{x} = B_O\mathbf{e}_1$, $\mathbf{x}' = B_O(c\mathbf{e}_1 + \sqrt{1-c^2}\mathbf{e}_2)$ one loses $o^2c^2(1-c^2)$ in $\|\operatorname{triu}\mathbf{D}\|^2$.

(viii) Lemma 3.3(a),(b) hold verbatim for $\Phi^{\neg yy}$. Without the quartic term, $\Upsilon^{\neg yy}$ is a positive semidefinite quadratic form in (y, y') plus a constant, hence convex, and its maximum over the square is at a vertex; by (d), $y = b$, $t = y' \in \{b, -b\}$. Now $F^{\neg yy}(s, -b) = 2o^2 + 2ro + 2ow + 4rw - 2s^2 - 2(r-w)s$ and $F^{\neg yy}(s, b) = 2o^2 - 2s^2 + 2(r+w)(o-s)$. Maximizing each over $|s| \le o$:

- $t = -b$: $\frac{1}{2}(2o + r + w)^2 + 2rw$ if $|r - w| \le 2o$; $4r(o+w)$ if $r - w > 2o$; $4w(o+r)$ if $w - r > 2o$.
- $t = b$: $\frac{1}{2}(2o + r + w)^2$ if $r + w \le 2o$, otherwise $4o(r+w)$.

Since $\frac{1}{2}(2o + r + w)^2 \ge 4o(r+w)$ (AM–GM), $4r(o+w) - 4o(r+w) = 4w(r-o) \ge 0$ in regime 2, and $4w(o+r) - 4o(r+w) = 4r(w-o) > 0$ in regime 3, the $t = -b$ values win, which gives the stated $\Delta_{\neg yy}$. Δ_G is the case $w = 0$: regime 1 if $r \le 2o$, regime 2 otherwise. $\Delta_y = \sup|y^2 - y'^2| = w$, attained at $(y, y') = (b, 0)$.

(ix) In regimes 1–2 the maximizer has $y' = -y$, so the $\mathbf{y}^\top\mathbf{y}$ term vanishes there and $\Delta = \Delta_{\neg yy}$ by the explicit formulas. In regime 3, $\Delta^2_{\neg yy} = 4w(r+o) = G(-b) \le \Delta^2$, so freeness is equivalent to $\max G \le 4w(r+o)$. By homogeneity, $F(s,t;r,o,w) = w^2F(s/w, t/b; r/w, o/w, 1)$, so take $w = b = 1$. Put $s = o - \varsigma$ with $\varsigma \in [0, 2o]$, $t = u - 1$ with $u \in [0, 2]$, $S = r + o$ and $\delta = 1 - r - 2o > 0$ (regime 3). A direct expansion (verified exactly in rational arithmetic by `s1_symbolic.py`, as are the other polynomial identities of this proof) gives

$$F(s,t) - 4S = 2\varsigma(u - \delta) - 2\varsigma^2 - S\,u(4-u) + u^2(2-u)^2. \tag{5}$$

Maximizing over $\varsigma \geq 0$ gives at most $\frac{1}{2}(u-\delta)_+^2 \leq \frac{1}{2}u^2$. If $w \leq 2(r+o)$, i.e. $S \geq \frac{1}{2}$, then $-Su(4-u) \leq -\frac{1}{2}u(4-u)$, and

$$F(s,t) - 4S \leq \tfrac{1}{2}u^2 - \tfrac{1}{2}u(4-u) + u^2(2-u)^2 = u(u-2)(u-1)^2 \leq 0 \qquad (0 \leq u \leq 2).$$

Hence $\Delta^2 \leq 4w(r+o) = \Delta^2_{\neg yy}$. If $w > 3(r+o)$, then $G(0) \geq F(o,0) = w(r+o+w) > 4w(r+o)$. For $o = 0$, $G(t) = r(b-t)^2 + (w-t^2)^2$, which is the function h of Proposition 3.5(d) with o replaced by w; the threshold follows from that computation. □

Remark 3.4 (The free set, numerically)**.** For fixed (r,o) the free set in regime 3 is an interval $r + 2o < w \leq w^*(r,o)$ [numerical: checked on a fine grid in w for $o/r \in [0,100]$]; Theorem 3.1(ix) proves $2(r+o) \leq w^* \leq 3(r+o)$ and $w^*(r,0) = \frac{11+5\sqrt{5}}{8}r$. The ratio $w^*/(r+o)$ decreases from 2.7725 at $o = 0$ to 2 as $o/r \to \infty$:

o/r	0	0.01	0.1	0.25	0.5	1	2	5	10	100
$w^*/(r+o)$	2.773	2.767	2.725	2.677	2.623	2.534	2.414	2.250	2.153	2.019

Example: $(B_R, B_O, B_y) = (1,1,5)$ has $w = 25 > 3(r+o) = 6$, so $\mathbf{y}^\top\mathbf{y}$ costs: $\Delta = 26.0706$ (maximizer $t^* = -0.0704$, $s^* = -0.324$), against $\Delta_{\neg yy} = \sqrt{200} = 14.142$. For $(\sqrt{5}, \sqrt{5}, 5)$, $w = 25 \leq w^* = 25.33$, so $\mathbf{y}^\top\mathbf{y}$ is free ($\Delta = \Delta_{\neg yy} = 31.62$), which is why the earlier paper saw no difference there. The indicative bounds $(5.59, 6.24, 2.89)$ are in regime 1 (free), with $\Delta = 86.15$ and $\Delta_G = 77.16$.[1]

3.5 The case $d_O = 1$

Proposition 3.5 ($d_O = 1$)**.** *Let $d_O = 1$ and $\Phi_1(x,x',y,y') = (x^2 - x'^2)^2 + r(x-x')^2 + r(y-y')^2 + (xy - x'y')^2 + (y^2 - y'^2)^2$ (here* triu $\mathbf{D} = \mathbf{D}$*, so there is no* triu *issue).*

- *(a) $\Delta^{(1)} \leq \Delta$, $\Delta^{(1)}_{\neg yy} \leq \Delta_{\neg yy}$ and $\Delta^{(1)}_G \leq \Delta_G$; so the $d_O \geq 2$ formulas are valid (conservative) for $d_O = 1$.*
- *(b) $\Delta^{(1)2} = \max\{\max_{x',t}\Phi_1(B_O,x',b,t),\ \max_{x',t}\Phi_1(B_O,x',t,b)\}$ over $(x',t) \in [-B_O,B_O]\times[-b,b]$.*
- *(c) $\Delta^{(1)} = \Delta$ in regime 2 and whenever some maximizer of Theorem 3.1 has $|s^\dagger| = o$ (for instance in the free part of regime 3, where the maximizer is $\mathbf{x}' = \mathbf{x}$, $y' = -y$). $\Delta^{(1)} < \Delta$ in the interior of regime 1.*
- *(d) $\Delta^{(1)2}_G = 4ro$ if $o \leq \frac{11+5\sqrt{5}}{8}r$; otherwise $\Delta^{(1)2}_G = \frac{o^2}{16}(3-q)^3(1+q)$ with $q = \sqrt{1-2r/o}$.*

Proof. (a) A $d_O = 1$ configuration embedded in $\mathbb{R}^2$ (second coordinates zero) has diagonal $\mathbf{D}$ and the same value. (b) Φ_1 is non-decreasing under $(x,x') \mapsto \kappa(x,x')$, $\kappa \geq 1$ (each x-dependent term is a non-negative form of degree 4 or 2 in (x,x')), and under $(y,y') \mapsto \kappa(y,y')$ (as in Lemma 3.3(c), with $\mathcal{Q} = r(y-y')^2 + (xy-x'y')^2$, a sum of squares). So $\max(|x|,|x'|) = B_O$ and $\max(|y|,|y'|) = b$. Using the symmetries $(x,y) \leftrightarrow (x',y')$, $(x,x') \mapsto -(x,x')$ and $(y,y') \mapsto -(y,y')$, take $x = B_O$; then either $y = b$ or $y' = b$. (c) In regime 2 the maximizer $\mathbf{x}' = -\mathbf{x}$ is collinear. In the interior of regime 1 every maximizer of the $d_O \geq 2$ problem has $u = v = o$ and $s = (w-r)/2 \in (-o,o)$, so it is not collinear. The $d_O = 1$ feasible set is a compact subset of the $d_O = 2$ set that contains no maximizer, so its maximum is strictly smaller. (d) With $w = 0$, $x = B_O$, $x' = B_O\tau$: $\Phi_1 = o\,h(\tau)$ with $h(\tau) = o(1-\tau^2)^2 + r(1-\tau)^2$, and $h'(\tau) = -2(1-\tau)\big[2o\tau(1+\tau) + r\big]$. If $o \leq 2r$ the bracket is $\geq r - o/2 \geq 0$, so h decreases and the maximum is $h(-1) = 4r$. If $o > 2r$, h has a local maximum at $\tau_+ = (q-1)/2$, where $h(\tau_+) = (1-\tau_+)^2[o(1+\tau_+)^2 + r] = \frac{o}{16}(3-q)^3(1+q)$, while

[1] The specification prints $\Delta = 86.18$ and $\Delta_G = 77.20$, presumably computed from unrounded bounds; with the bounds as printed the formulas give 86.152 and 77.162.

$h(-1) = 4r = 2o(1-q^2)$. The interior value wins iff $(3-q)^3 > 32(1-q)$, i.e. $q > q_0 = 5 - 2\sqrt{5}$, i.e. $o/r > 2/(1-q_0^2) = \frac{11+5\sqrt{5}}{8}$ (and $o \le 2r$ falls in the first case). □

For $d_O = 1$ we give no closed form for $\Delta^{(1)}$ in regimes 1 and 3. (b) is an exact two-variable polynomial maximization over a rectangle, solvable to machine precision; for calibration the $d_O \ge 2$ formula is a valid upper bound. The losses are substantial in regime 1: $\Delta^{(1)}/\Delta = 0.894$ at $(1,1,1)$, 0.784 at $(1,2,1)$ and 0.912 at the indicative bounds (Table 2).

3.6 Numerical verification

Script `scratch_stats/s1_check.py` (output `s1_check.out`) compares, for 14 bound settings covering all regimes: the exact value of Theorem 3.1 (closed form, or candidate enumeration in regime 3); `sim_core.delta_rep` (read only); a brute-force multistart BFGS maximization of the triu objective Φ_{triu} over raw configurations $(\mathbf{x}, \mathbf{x}', y, y')$ for $d_O = 2$ and $d_O = 3$; and the explicit worst-case configuration of (vii) evaluated with the triu norm. All agree to 10^{-6} (relative difference to `delta_rep` is $\le 10^{-9}$, which is its deliberate rounding up). $\Delta_{\neg yy}$ agrees with brute force, and the $d_O = 1$ reduction (b) agrees with brute force over $d_O = 1$ configurations. Script `s1_freeset.py` produced the table of Remark 3.4, and `s1_band.py` confirms $\max_t G - 4(r+o) \le 4.4 \times 10^{-16}$ over a 300×120 grid of the band $r + 2o < w \le 2(r+o)$.

B_R	B_O	B_y	reg.	Δ exact	brute d_O=2	$\Delta_{\neg yy}$	$\Delta^{(1)}$ (d_O=1)	Δ_G	$\Delta_G^{(1)}$
1	1	1	1	3.162278	3.162278	3.16228	2.82843	2.12132	2.00000
1	1	5	3	26.070598	26.070598	14.14214	26.05763	2.12132	2.00000
3	1	1	2	8.485281	8.485281	8.48528	8.48528	6.00000	6.00000
2.236	2.236	5	3	31.621815	31.621815	31.62182	31.62182	10.60596	9.99939
5.59	6.24	2.89	1	86.152245	86.152245	86.15224	78.53500	77.16183	69.76320
1	2	1	1	7.211103	7.211103	7.21110	5.65685	6.36396	4.53635
4	1	0.5	2	8.944272	8.944272	8.94427	8.94427	8.00000	8.00000
1	0.5	3	3	9.672258	9.672258	6.70820	9.67226	1.00000	1.00000
1.5	0.7	2.2	3	7.283296	7.283296	7.28330	7.28330	2.10000	2.10000
1	1	1.8	3	5.091169	5.091169	5.09117	5.09117	2.12132	2.00000
1	0.2	2	3	4.566806	4.566806	4.07922	4.56681	0.40000	0.40000
1	3	4	1	25.387005	25.387005	25.38700	25.29822	13.43503	9.51475

Table 2: Proposition S1: the exact value equals `delta_rep`, the brute-force triu maximum for $d_O = 2$ and $d_O = 3$ (the $d_O = 3$ column is identical and omitted), and the explicit configuration of Theorem 3.1(vii), in every case.

4 Lemma S2: operator norm of the masked noise

Lemma 4.1 (S2). *Let* $\mathbf{E}$ *be a masked symmetric noise matrix with scale* σ *(Section 2); the proof allows any mask and any variances* $\le \sigma^2$ *off the diagonal and* $\le 2\sigma^2$ *on it, so it covers the full GOE-type matrix too. Let* $\mathbf{g} \sim \mathcal{N}(0, \mathbf{I}_p)$.

(a) $\mathbb{E}\lambda_{\max}(\mathbf{E}) \le 2\sigma\,\mathbb{E}\|\mathbf{g}\| < 2\sigma\sqrt{p}$, *and* $\lambda_{\max}(-\mathbf{E})$ *has the same law as* $\lambda_{\max}(\mathbf{E})$.

(b) $\mathbf{E} \mapsto \lambda_{\max}(\mathbf{E})$ *and* $\mathbf{E} \mapsto \|\mathbf{E}\|_{\text{op}}$ *are* $\sqrt{2}\sigma$*-Lipschitz functions of the standard Gaussian vector of independent entries. Hence, for* $t \ge 0$,

$$\Pr[\lambda_{\max}(\mathbf{E}) \ge \mathbb{E}\lambda_{\max}(\mathbf{E}) + t] \le e^{-t^2/(4\sigma^2)}, \qquad \Pr[\|\mathbf{E}\|_{\text{op}} \ge \mathbb{E}\|\mathbf{E}\|_{\text{op}} + t] \le e^{-t^2/(4\sigma^2)}.$$

(c) $\Pr\big[\lambda_{\max}(\mathbf{E}) \ge \sigma\big(2\sqrt{p} + 2\sqrt{\ln(1/\alpha)}\big)\big] \le \alpha$ *and* $\Pr\big[\|\mathbf{E}\|_{\mathrm{op}} \ge \sigma\big(2\sqrt{p} + 2\sqrt{\ln(2/\alpha)}\big)\big] \le \alpha$.
(d) $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} \le 2\sigma\sqrt{p} + c_0\sigma$, *with* $c_0 = 2\sqrt{\ln 2} + 4\sqrt{\pi}\,Q(\sqrt{2\ln 2}) \approx 2.512$, *and* $(\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}}^4)^{1/4} \le \sigma(2\sqrt{p} + c_0 + 2\sqrt{2})$.

In Algorithm 6, $r_1 = \sigma_1(2\sqrt{p} + 2\sqrt{\ln(1/\alpha_L)})$ *therefore satisfies* $\Pr[\lambda_{\max}(\hat{\mathbf{G}}_1 - \mathbf{G}) > r_1] \le \alpha_L$, *and this one-sided event is all the algorithm uses (Lemma S3).*

Proof. (a) For a unit vector $\mathbf{u}$ let $X_{\mathbf{u}} = \mathbf{u}^\top\mathbf{E}\mathbf{u} = \sum_{i\le j,\chi_{ij}=1} E_{ij}\, w_{ij}(\mathbf{u})$, with $w_{ii} = u_i^2$ and $w_{ij} = 2u_iu_j$. Then

$$\mathbb{E}(X_{\mathbf{u}} - X_{\mathbf{v}})^2 \le \sigma^2\Big[2\sum_i (u_i^2 - v_i^2)^2 + 4\sum_{i<j}(u_iu_j - v_iv_j)^2\Big] = 2\sigma^2\|\mathbf{u}\mathbf{u}^\top - \mathbf{v}\mathbf{v}^\top\|_F^2$$
$$= 4\sigma^2\big(1 - (\mathbf{u}^\top\mathbf{v})^2\big) \le 4\sigma^2\|\mathbf{u} - \mathbf{v}\|^2,$$

using $1 - (\mathbf{u}^\top\mathbf{v})^2 = (1 - \mathbf{u}^\top\mathbf{v})(1 + \mathbf{u}^\top\mathbf{v}) \le 2(1 - \mathbf{u}^\top\mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|^2$. The process $Y_{\mathbf{u}} = 2\sigma\,\mathbf{g}^\top\mathbf{u}$ has $\mathbb{E}(Y_{\mathbf{u}} - Y_{\mathbf{v}})^2 = 4\sigma^2\|\mathbf{u} - \mathbf{v}\|^2$. By the Sudakov–Fernique inequality [12, 9] (see [14, Thm. 7.2.11]; both processes are centred, Gaussian and continuous on the sphere), $\mathbb{E}\lambda_{\max}(\mathbf{E}) = \mathbb{E}\sup_{\mathbf{u}} X_{\mathbf{u}} \le \mathbb{E}\sup_{\mathbf{u}} Y_{\mathbf{u}} = 2\sigma\mathbb{E}\|\mathbf{g}\| \le 2\sigma\sqrt{p}$. $\mathbf{E}$ and $-\mathbf{E}$ have the same law.

(b) Write $E_{ij} = \sigma_{ij}\zeta_{ij}$ with independent standard ζ_{ij}. For a unit $\mathbf{u}$, $|\mathbf{u}^\top(\mathbf{E}(\zeta) - \mathbf{E}(\zeta'))\mathbf{u}| \le \|\zeta - \zeta'\|\big(\sum \sigma_{ij}^2 w_{ij}(\mathbf{u})^2\big)^{1/2} \le \sqrt{2}\sigma\|\zeta - \zeta'\|$, since $\sum_i u_i^4 \cdot 2 + \sum_{i<j} 4u_i^2u_j^2 = 2$ (and with diagonal variance σ^2 the bound is $\sigma^2(2 - \sum_i u_i^4)$). Both $\lambda_{\max}$ and $\|\cdot\|_{\mathrm{op}} = \sup_{\mathbf{u}}|\mathbf{u}^\top\cdot\mathbf{u}|$ are therefore $\sqrt{2}\sigma$-Lipschitz, and Gaussian concentration [13] (see [5, Thm. 5.6]) gives $\Pr[f - \mathbb{E}f \ge t] \le e^{-t^2/(2L^2)}$ with $L = \sqrt{2}\sigma$.

(c) Combine (a) and (b) with $t = 2\sigma\sqrt{\ln(1/\alpha)}$. For the norm, $\|\mathbf{E}\|_{\mathrm{op}} = \max(\lambda_{\max}(\mathbf{E}), \lambda_{\max}(-\mathbf{E}))$ and a union bound gives the factor 2.

(d) With $m = 2\sigma\sqrt{p} \ge \mathbb{E}\lambda_{\max}(\pm\mathbf{E})$, $\|\mathbf{E}\|_{\mathrm{op}} \le m + \max\big((\lambda_{\max}(\mathbf{E}) - m)_+, (\lambda_{\max}(-\mathbf{E}) - m)_+\big)$, and the last maximum exceeds t with probability at most $\min(1, 2e^{-t^2/4\sigma^2})$; integrating gives c_0. For the fourth moment, $\|\mathbf{E}\|_{\mathrm{op}} = \mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} + X$ with $\Pr[|X| \ge t] \le 2e^{-t^2/(4\sigma^2)}$ by (b), so $\mathbb{E}X^4 \le 8\int_0^\infty t^3 e^{-t^2/(4\sigma^2)}dt = 64\sigma^4$, and Minkowski's inequality gives the claim. □

Remark 4.2 (What is not proved)**.** The claim $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} \le 2\sigma\sqrt{p}$, which would give the two-sided bound with $\ln(1/\alpha)$, is [open] here. Sudakov–Fernique applied to $\sup_{\mathbf{u},\pm} \pm\mathbf{u}^\top\mathbf{E}\mathbf{u}$ fails, because the cross-sign increments $\mathbb{E}(\mathbf{u}^\top\mathbf{E}\mathbf{u} + \mathbf{v}^\top\mathbf{E}\mathbf{v})^2$ reach $8\sigma^2$ at $\mathbf{u} = \mathbf{v}$. The suggested Gaussian comparison does reduce the masked case to the full one: if $\mathbf{W} = \mathbf{E} + \mathbf{E}'$ with $\mathbf{E}'$ an independent centred Gaussian symmetric matrix supported on the masked entries (and, if desired, extra diagonal variance up to $2\sigma^2$), then $\mathbb{E}\|\mathbf{W}\|_{\mathrm{op}} = \mathbb{E}_{\mathbf{E}}\mathbb{E}_{\mathbf{E}'}\|\mathbf{E} + \mathbf{E}'\|_{\mathrm{op}} \ge \mathbb{E}\|\mathbf{E}\|_{\mathrm{op}}$ by Jensen's inequality, since $\|\cdot\|_{\mathrm{op}}$ is convex and $\mathbb{E}\mathbf{E}' = 0$. So the mask can only help, and it would suffice to prove $\mathbb{E}\|\mathbf{W}\|_{\mathrm{op}} \le 2\sigma\sqrt{p}$ for the full GOE-type matrix. For the GOE this holds asymptotically (the largest eigenvalue has mean $2\sigma\sqrt{p} - O(\sigma p^{-1/6})$ by Tracy–Widom asymptotics), but I know of no non-asymptotic proof with the constant exactly 2. Known sharp non-asymptotic bounds, such as that of Bandeira and van Handel [4], carry an additional $O(\sigma\sqrt{\log p})$ term. Monte Carlo (`s2_goe_ratio.py`) gives $\mathbb{E}\|\mathbf{W}\|_{\mathrm{op}}/(2\sigma\sqrt{p}) = 0.567, 0.724, 0.799, 0.864, 0.926, 0.954, 0.966$ for $p = 1, 2, 3, 5, 11, 20, 30$ (GOE normalization), increasing towards 1 from below, and 0.81 for the protocol's masked $p = 11$ (Table 3). Algorithm 6 does not need the claim.

Subgaussian (discrete-Gaussian) noise. By Lemma 2.2(b),(c), (c) holds for the protocol's noise with α replaced by $\alpha + N\,\mathrm{TV}_1(\sigma_{c,1}) + \delta_B$ and r_1 by $r_1 + \sqrt{2}\,\bar{\nu}$. Without the coupling, using only that the noise vector is σ^2-subgaussian (Lemma 2.2(a)), a $\frac{1}{4}$-net $\mathcal{N}$ of the sphere ($|\mathcal{N}| \le 9^p$)

and $\|\mathbf{E}\|_{\mathrm{op}} \le 2\max_{\mathbf{v}\in\mathcal{N}}|\mathbf{v}^\top\mathbf{E}\mathbf{v}|$ (see [14, Cor. 4.2.13, Ex. 4.4.3]), together with the fact that $\mathbf{v}^\top\mathbf{E}\mathbf{v} = \langle \mathbf{w}(\mathbf{v}), \boldsymbol{\xi}\rangle$ with $\|\mathbf{w}(\mathbf{v})\|^2 \le 2$ is $2\sigma^2$-subgaussian, give

$$\Pr\Big[\|\mathbf{E}\|_{\mathrm{op}} \ge 4\sigma\sqrt{p\ln 9 + \ln(2/\alpha)}\Big] \le \alpha.$$

For $p = 11$ and $\alpha = 0.0025$ this is 22.2σ, against 11.5σ for the Gaussian r_1. The coupling route is therefore the one to use.

Monte Carlo (`s2_check.py`, 2×10^5 draws). For the protocol shape ($p = 11$, 5×5 exact block): $\mathbb{E}\lambda_{\max} = 4.98\sigma$, $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} = 5.38\sigma$, against $2\sigma\sqrt{p} = 6.63\sigma$, a ratio of 0.81. The 0.9975-quantile of $\lambda_{\max}$ is 7.43σ (of $\|\mathbf{E}\|_{\mathrm{op}}$, 7.62σ), while $r_1/\sigma_1 = 11.53$ at $\alpha_L = 0.0025$, so r_1 is conservative by a factor of about 1.55. Table 3 gives other shapes. The discrete-Gaussian version (ring $N = 128$, canonical embedding, $\sigma_c \in \{0.8, 3\}$) has exactly uncorrelated slots and the same $\lambda_{\max}$ distribution to Monte Carlo accuracy (mean 4.97, 0.9975-quantile 7.38).

shape	$2\sqrt{p}$	$\mathbb{E}\lambda_{\max}$	$\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}}$	$q_{.9975}(\lambda_{\max})$	r_1/σ_1	$\Pr[\lambda_{\max} \ge 2\sqrt{p}+2]$
$p = 11$, $d_A = 5$ (protocol)	6.633	4.976	5.383	7.43	11.53	2.5×10^{-5}
$p = 11$, no mask, diag var 1	6.633	5.431	5.858	7.89	11.53	1.8×10^{-4}
$p = 11$, GOE (diag var 2)	6.633	5.645	6.148	8.29	11.53	9.0×10^{-4}
$p = 2$, $d_A = 1$	2.828	0.965	1.368	3.60	7.72	9×10^{-5}
$p = 20$, $d_A = 5$	8.944	7.702	8.096	10.00	13.84	4×10^{-5}
$p = 50$, $d_A = 10$	14.142	13.095	13.446	15.09	19.04	0

Table 3: Lemma S2, Monte Carlo in units of σ; r_1/σ_1 at $\alpha_L = 0.0025$. The bound $e^{-t^2/4}$ at $t = 2$ is 0.37; the observed tails are far lighter, because the Lipschitz constant $\sqrt{2}\sigma$ is attained only by spread-out eigenvectors (empirical s.d. of $\lambda_{\max}$: 0.77σ).

5 Lemma S3: the two-stage choice of the ridge

Let $\mathbf{E}_1 = \hat{\mathbf{G}}_1 - \mathbf{G}$ (scale σ_1), $\gamma := g^\star + r_1$ with $g^\star = 2\vartheta^\star\sigma_2\sqrt{p}$, and

$$\Omega_L := \{\lambda_{\max}(\mathbf{E}_1) \le r_1\}.$$

Lemma 5.1 (S3). *Under Assumption 2.1:*

(a) ***Measurability and independence.*** $(\lambda, \Psi) = \Lambda(\mathbf{A}, \hat{\mathbf{B}}_1, \hat{\mathbf{C}}_1)$ *for a Borel map* Λ *that depends otherwise only on public constants* $(\sigma_1, \sigma_2, p, \vartheta^\star, \alpha_L, \kappa_\Psi)$. *Hence, conditionally on the data* $(\mathbf{X}, \mathbf{y})$, $(\lambda, \Psi, \hat{L})$ *is independent of the stage-2 noise* $(\mathbf{E}_2, \mathbf{f}_2, g)$ *and of the separate* $\mathbf{y}^\top\mathbf{y}$ *noise.*

(b) ***Radius event.*** $\Pr[\Omega_L \mid \mathbf{X}, \mathbf{y}] \ge 1 - \alpha_L$. *On* Ω_L: $\lambda_{\min}(\mathbf{G} + \lambda\Psi) \ge g^\star$, *i.e.* $\vartheta_0 \ge \vartheta^\star$; *and* $\hat{L} \le \lambda_{\min}(\mathbf{G})$.

(c) ***The*** Π_O ***ridge.*** $\lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda\Pi_O) \le \lambda_{\min}(\mathbf{A})$ *for every* $\lambda \ge 0$ *(Cauchy interlacing), and* $\lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda\Pi_O) \uparrow \lambda_{\min}(\mathbf{A})$ *as* $\lambda \to \infty$. *If* $\lambda_{\min}(\mathbf{A}) > \gamma$,

$$\lambda_O = \max\big\{0,\ \gamma - \lambda_{\min}\big(\hat{\mathbf{B}}_1 - \hat{\mathbf{C}}_1^\top(\mathbf{A} - \gamma\mathbf{I})^{-1}\hat{\mathbf{C}}_1\big)\big\} < \infty;$$

if $\lambda_{\min}(\mathbf{A}) < \gamma$, $\lambda_O = \infty$; *the case* $\lambda_{\min}(\mathbf{A}) = \gamma$ *has probability zero (set* $\lambda_O = \infty$*). No bisection is needed.*

(d) ***Gate.*** *Conditionally on the data and stage 1,* $\Pr[\hat{\vartheta} < 1] \le \Pr\big[\lambda_{\max}(\mathbf{E}_2) > 2(\vartheta_0 - 1)\sigma_2\sqrt{p}\big] \le \exp\big(-p(\vartheta_0 - 2)^2\big)$ *for* $\vartheta_0 > 2$. *Hence* $\Pr[\textsf{refused} \mid \mathbf{X}, \mathbf{y}] \le \alpha_L + \exp(-p(\vartheta^\star - 2)^2)$ *when* $\vartheta^\star > 2$.

Proof. (a) $\lambda_{\min}$ is continuous; $\hat{L}$, λ_I and the closed form in (c) are continuous on their domains, and the comparison $\lambda_O \le \kappa_\Psi \lambda_I$ is a Borel event. Equivalently, $\{\lambda_O \le a\} = \{\lambda_{\min}(\hat{\mathbf{G}}_1 + a\Pi_O) \ge \gamma\}$ is closed, because $\lambda \mapsto \lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda\Pi_O)$ is continuous and non-decreasing; so λ_O is lower semicontinuous. A bisection with a fixed number of steps is also Borel. The stage-2 noise is fresh: η and (u, e_0, e_1) are drawn anew in Algorithm 5, and (λ, Ψ) is a function of the data and $\mathbf{E}_1$ only. For the real protocol this independence holds exactly for the ideal mechanism $\mathcal{M}$ and up to the perturbation of norm $\le \sqrt{2}\,\bar{\nu}$ of Lemma 2.2.

(b) $\Pr[\Omega_L] \ge 1 - \alpha_L$ is Lemma 4.1(c). For any symmetric $\mathbf{M}$, $\lambda_{\min}(\mathbf{G}+\mathbf{M}) = \lambda_{\min}(\hat{\mathbf{G}}_1 + \mathbf{M} - \mathbf{E}_1) \ge \lambda_{\min}(\hat{\mathbf{G}}_1 + \mathbf{M}) - \lambda_{\max}(\mathbf{E}_1)$ (Weyl). With $\mathbf{M} = \lambda\Psi$: if $\Psi = \mathbf{I}$, $\lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda_I \mathbf{I}) = \lambda_{\min}(\hat{\mathbf{G}}_1) + \lambda_I \ge \gamma$; if $\Psi = \Pi_O$ (then $\lambda_O < \infty$), $\lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda_O \Pi_O) \ge \gamma$ by definition. So on Ω_L, $\lambda_{\min}(\mathbf{G}_\lambda) \ge \gamma - r_1 = g^\star$. With $\mathbf{M} = 0$: $\lambda_{\min}(\mathbf{G}) \ge \lambda_{\min}(\hat{\mathbf{G}}_1) - r_1$, and $\lambda_{\min}(\mathbf{G}) \ge 0$, so $\lambda_{\min}(\mathbf{G}) \ge \hat{L}$.

(c) Interlacing: $\mathbf{A}$ is a principal submatrix of $\hat{\mathbf{G}}_1 + \lambda\Pi_O$ (the $\mathbf{A}$ block is exact and unaffected by $\lambda\Pi_O$). For $\gamma < \lambda_{\min}(\mathbf{A})$, $\lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda\Pi_O) \ge \gamma$ iff $\left(\begin{smallmatrix} \mathbf{A}-\gamma\mathbf{I} & \hat{\mathbf{C}}_1 \\ \hat{\mathbf{C}}_1^\top & \hat{\mathbf{B}}_1 + (\lambda-\gamma)\mathbf{I} \end{smallmatrix}\right) \succeq 0$ iff (Schur complement, $\mathbf{A} - \gamma\mathbf{I} \succ 0$) $\hat{\mathbf{B}}_1 - \hat{\mathbf{C}}_1^\top (\mathbf{A} - \gamma\mathbf{I})^{-1} \hat{\mathbf{C}}_1 + (\lambda - \gamma)\mathbf{I} \succeq 0$. The limit follows from the Schur complement as $\lambda \to \infty$. If $\gamma = \lambda_{\min}(\mathbf{A})$, feasibility requires $\hat{\mathbf{C}}_1^\top \mathbf{v} = 0$ for an eigenvector $\mathbf{v}$ of $\mathbf{A}$, a null event under continuous noise.

(d) $\lambda_{\min}(\hat{\mathbf{G}}_2 + \lambda\Psi) \ge \lambda_{\min}(\mathbf{G}_\lambda) - \lambda_{\max}(-\mathbf{E}_2)$, so $\hat{\vartheta} < 1$ implies $\lambda_{\max}(-\mathbf{E}_2) > 2(\vartheta_0 - 1)\sigma_2\sqrt{p} = 2\sigma_2\sqrt{p} + 2(\vartheta_0 - 2)\sigma_2\sqrt{p}$. Apply Lemma 4.1(a),(b) with $t = 2(\vartheta_0 - 2)\sigma_2\sqrt{p}$ and (b) of this lemma. □

Remark 5.2 (Size of the guarantees)**.** With $\vartheta^\star = 2.5$ and $p = 11$ the rigorous refusal bound is $\alpha_L + e^{-2.75} = \alpha_L + 0.064$. Monte Carlo at the worst case $\vartheta_0 = \vartheta^\star$ found no exceedance of $\lambda_{\max}(\mathbf{E}) > 9.95\sigma$ in 2×10^5 draws, and Algorithm 6 never refused in our simulations (Table 5). Because r_1 is conservative (Lemma S2), the realized ϑ_0 is typically far above $\vartheta^\star$: it was 7.1–8.2 in the ridge scenarios of Table 4. The private bound $\sqrt{U_{yy}/\hat{L}}$ is finite only if $\lambda_{\min}(\hat{\mathbf{G}}_1) > r_1$; when the ridge is active this often fails, and the β interval then rests on B_β^{pub} (Correction 10).

6 Lemma S4: the bound on $\|\boldsymbol{\beta}\|$

Lemma 6.1 (S4)**.** *Let* $\mathbf{X}$ *be the matched (clipped) design,* $\mathbf{G} = \mathbf{X}^\top\mathbf{X}$ *with* $\lambda_{\min}(\mathbf{G}) > 0$*, and* $\mathbf{y}$ *the clipped response,* $|y_i| \le B_y$*.*

(a) If $\mathbb{E}[\mathbf{y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta}$*, then* $\|\boldsymbol{\beta}\|^2 \le \boldsymbol{\beta}^\top\mathbf{G}\boldsymbol{\beta}/\lambda_{\min}(\mathbf{G}) \le \mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}]/\lambda_{\min}(\mathbf{G})$*. Without any model, the projection coefficient* $\boldsymbol{\beta}^* := \mathbf{G}^{-1}\mathbf{X}^\top\mathbb{E}[\mathbf{y} \mid \mathbf{X}]$ *satisfies the same bound.*

(b) If the y_i*,* $i \in I$*, are conditionally independent given* $\mathbf{X}$*, then for* $s \ge 0$

$$\Pr\big[\widehat{yy} \le \mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}] - s \;\big|\; \mathbf{X}\big] \le \exp\Big(-\frac{s^2}{2(nB_y^4/4 + \sigma_{yy}^2)}\Big).$$

Hence $U_{yy}^{\text{c}} := \widehat{yy} + \sqrt{2\ln(1/\alpha_U)\,(nB_y^4/4 + \sigma_{yy}^2)}$ *satisfies* $\Pr[U_{yy}^{\text{c}} \ge \mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}] \mid \mathbf{X}] \ge 1 - \alpha_U$*. The specification's* $U_{yy} = \widehat{yy} + z_{1-\alpha_U}\sigma_{yy} + B_y^2\sqrt{n\ln(1/\alpha_U)/2}$ *satisfies this with probability* $\ge 1 - 2\alpha_U$ *only; with* $\alpha_U/2$ *in both terms it satisfies it with probability* $\ge 1 - \alpha_U$*. In all cases one may replace* U *by* $\min\{U, nB_y^2\}$*, since* $\mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}] \le nB_y^2$ *deterministically.*

(c) Let $\Omega_U = \{U \ge \mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}]\}$*. On* $\Omega_L \cap \Omega_U$*, and if* $\|\boldsymbol{\beta}\| \le B_\beta^{\text{pub}}$*, we have* $\|\boldsymbol{\beta}\| \le \hat{B}_\beta = \min\{B_\beta^{\text{pub}}, \sqrt{U/\hat{L}}\}$*.*

Proof. (a) $\mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}] = \|\mathbb{E}[\mathbf{y} \mid \mathbf{X}]\|^2 + \sum_i \mathrm{Var}(y_i \mid \mathbf{X}) \ge \|\mathbb{E}[\mathbf{y} \mid \mathbf{X}]\|^2$. Under the model, $\|\mathbb{E}[\mathbf{y} \mid \mathbf{X}]\|^2 = \boldsymbol{\beta}^\top\mathbf{G}\boldsymbol{\beta} \ge \lambda_{\min}(\mathbf{G})\|\boldsymbol{\beta}\|^2$. In general, $\boldsymbol{\beta}^{*\top}\mathbf{G}\boldsymbol{\beta}^* = \|\mathbf{H}\mathbb{E}[\mathbf{y} \mid \mathbf{X}]\|^2 \le \|\mathbb{E}[\mathbf{y} \mid \mathbf{X}]\|^2$ with $\mathbf{H}$ the hat matrix.

(b) $y_i^2 \in [0, w] = [0, B_y^2]$, so by Hoeffding's lemma [10] each centred y_i^2 is $(w^2/4) = (B_y^4/4)$-subgaussian given $\mathbf{X}$; independence gives $nw^2/4$ for the sum, and adding the independent $\mathcal{N}(0, \sigma_{yy}^2)$ release noise gives $nw^2/4 + \sigma_{yy}^2$. The Chernoff bound gives the display. For the specification's form, Hoeffding gives $\Pr[\mathbf{y}^\top\mathbf{y} \le \mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}] - w\sqrt{n\ln(1/\alpha)/2}] \le \alpha$ (this term is correct as stated) and $\Pr[\widehat{yy} \le \mathbf{y}^\top\mathbf{y} - z_{1-\alpha}\sigma_{yy}] = \alpha$; the union bound gives 2α, and $\alpha/2$ in each restores α. Without conditional independence, Hoeffding does not apply; a martingale version controls $\sum_i (y_i^2 - \mathbb{E}[y_i^2 \mid \mathcal{F}_{i-1}])$, which is not the target, so only the deterministic bound nB_y^2 remains.

(c) On Ω_L, $\hat{L} \le \lambda_{\min}(\mathbf{G})$ (Lemma 5.1(b)); on Ω_U, $U \ge \mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}]$; combine with (a). If $\hat{L} = 0$ the private bound is $+\infty$ and the claim reduces to the public one. □

Clipping and the estimand. R clips $\mathbf{x}_R$, O clips $\mathbf{x}_O$ and y. The clipped data need not follow a linear model even if the raw data do. Clipping y is a non-linear map of $\mathbf{x}^\top\boldsymbol{\beta}+\varepsilon$, and clipping $\mathbf{x}$ breaks linearity in the original features. Clipping also makes the errors heteroskedastic: $\mathrm{Var}(\mathrm{clip}(y) \mid \mathbf{x})$ depends on $\mathbf{x}$ through the probability of clipping. Precisely:

- The Wald interval targets $\boldsymbol{\theta}_\lambda = (\mathbf{G} + \lambda\Psi)^{-1}\mathbf{c}$ computed on the clipped matched data. It is well defined without any model; its randomness is the mechanism only, given the data and stage 1.
- The bias-aware interval targets $\boldsymbol{\beta}$ in the *working model on the clipped scale*: $\mathbb{E}[\tilde{y}_i \mid \tilde{\mathbf{X}}] = \tilde{\mathbf{x}}_i^\top\boldsymbol{\beta}$ with conditionally independent, homoskedastic errors, where tildes denote clipped values. Without that model the natural target is the conditional projection $\boldsymbol{\beta}^* = \mathbf{G}^{-1}\tilde{\mathbf{X}}^\top\mathbb{E}[\tilde{\mathbf{y}} \mid \tilde{\mathbf{X}}]$. For $\boldsymbol{\beta}^*$ the bias identity $\mathbb{E}[\boldsymbol{\theta}_\lambda \mid \mathbf{X}] - \boldsymbol{\beta}^* = -\lambda\mathbf{P}_\lambda\Psi\boldsymbol{\beta}^*$ (because $\tilde{\mathbf{X}}^\top\mathbb{E}[\tilde{\mathbf{y}} \mid \tilde{\mathbf{X}}] = \mathbf{G}\boldsymbol{\beta}^*$) and Lemma 6.1(a) still hold, but the homoskedastic variance formula does not. The interval is then approximate, and a sandwich variance (a further release) would be needed for validity.
- The raw-scale coefficient equals the clipped-scale one only if clipping never binds on the support, i.e. if the bounds are true bounds. Exactly Gaussian errors are incompatible with $|y| \le B_y$, so normality of the sampling term can only be asymptotic (Section 7).

7 Proposition S5: coverage

Condition on the data and stage 1, $\mathcal{F}_1 = \sigma(\mathbf{X}, \mathbf{y}, \mathbf{E}_1)$, so $\lambda, \Psi, \mathbf{G}_\lambda, \mathbf{P}_\lambda, \boldsymbol{\theta}_\lambda, \vartheta_0$ are fixed. Let

$$\boldsymbol{\xi} := \mathbf{P}_\lambda(\mathbf{f}_2 - \mathbf{E}_2\boldsymbol{\theta}_\lambda), \qquad \mathbf{V}_0 := \mathrm{Var}(\boldsymbol{\xi} \mid \mathcal{F}_1) = \sigma_2^2\,\mathbf{P}_\lambda\big(\mathbf{I} + \mathbf{K}(\boldsymbol{\theta}_\lambda)\big)\mathbf{P}_\lambda, \qquad q := \|\mathbf{G}_\lambda^{-1/2}\mathbf{E}_2\mathbf{G}_\lambda^{-1/2}\|_{\mathrm{op}}.$$

Theorem 7.1 (S5(a): Wald interval for $\boldsymbol{\theta}_\lambda$)**.** *Under Assumption 2.1:*

(i) *[exact]* $\boldsymbol{\xi} \mid \mathcal{F}_1 \sim \mathcal{N}(0, \mathbf{V}_0)$*; and on* $\{\mathbf{G}_\lambda + \mathbf{E}_2$ *invertible*$\}$, $\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_\lambda = (\mathbf{I} + \mathbf{P}_\lambda\mathbf{E}_2)^{-1}\boldsymbol{\xi}$.

(ii) *[exact] If* $q < 1$*, then* $|\tilde{\theta}_j - \theta_{\lambda,j} - \xi_j| \le \frac{q}{1-q}\sqrt{(\mathbf{P}_\lambda)_{jj}}\,\|\mathbf{G}_\lambda^{1/2}\boldsymbol{\xi}\|$.

(iii) *[exact, finite sample] Let* $s_j^2 = (\mathbf{V}_0)_{jj}$, $z = z_{1-\alpha/2}$, $\varpi \in (0, z)$ *and* $\eta \in (0, 1)$ *with* $\eta\vartheta_0 \ge 1$*. The oracle interval* $\tilde{\theta}_j \pm z\,s_j$ *satisfies*

$$\Pr\big[|\tilde{\theta}_j - \theta_{\lambda,j}| > z s_j \;\big|\; \mathcal{F}_1\big] \le \alpha + 2\varpi\,\phi(z - \varpi) + 2e^{-p(\eta\vartheta_0 - 1)^2} + \Big(\frac{\eta}{1-\eta}\Big)^2\frac{\omega_j}{\varpi^2},$$

with $\omega_j := (\mathbf{P}_\lambda)_{jj}\,\sigma_2^2\,\mathrm{tr}\big((\mathbf{I} + \mathbf{K}(\boldsymbol{\theta}_\lambda))\mathbf{P}_\lambda\big)/s_j^2 \le (1 + 2\|\boldsymbol{\theta}_\lambda\|^2)\,\mathrm{tr}(\mathbf{P}_\lambda)/(\mathbf{P}_\lambda)_{jj}$.

(iv) *[first order] The reported interval uses* $\mathbf{V}_{\mathrm{mech}}$ *in place of* $\mathbf{V}_0$*. By Lemma S6,* $(\mathbf{V}_{\mathrm{mech}})_{jj} = s_j^2(1 + O_p(\vartheta_0^{-1}) + O(\vartheta_0^{-2}) + O_p(B^{-1/2}))$.

(v) *[asymptotic] Unconditionally,* $\Pr[\theta_{\lambda,j} \notin \mathrm{CI}_j,$ *not refused*$] \le \alpha_{\mathrm{CI}} + \alpha_L + \epsilon_j$*, where* $\epsilon_j \to 0$ *as* $\vartheta^\star \to \infty$ *and* $B \to \infty$ *with* p, $\|\boldsymbol{\theta}_\lambda\|$ *and* $\mathrm{tr}(\mathbf{P}_\lambda)/(\mathbf{P}_\lambda)_{jj}$ *bounded (choose* $\eta = c/\vartheta_0$ *and* $\varpi \to 0$ *slowly in (iii)).*

Proof. (i) Given $\mathcal{F}_1$, $(\mathbf{E}_2, \mathbf{f}_2)$ is Gaussian, centred and independent of (λ, Ψ) (Lemma 5.1(a)), and $\mathbf{E}_2 \perp \mathbf{f}_2$; so $\boldsymbol{\xi}$ is a centred Gaussian linear image with covariance $\mathbf{P}_\lambda(\sigma_2^2\mathbf{I} + \sigma_2^2\mathbf{K})\mathbf{P}_\lambda$. From $(\mathbf{G}_\lambda+\mathbf{E}_2)\tilde{\boldsymbol{\theta}} = \mathbf{c}+\mathbf{f}_2$ and $\mathbf{G}_\lambda\boldsymbol{\theta}_\lambda = \mathbf{c}$ we get $(\mathbf{G}_\lambda+\mathbf{E}_2)(\tilde{\boldsymbol{\theta}}-\boldsymbol{\theta}_\lambda) = \mathbf{f}_2-\mathbf{E}_2\boldsymbol{\theta}_\lambda$. (ii) Let $\mathbf{M} = \mathbf{G}_\lambda^{-1/2}\mathbf{E}_2\mathbf{G}_\lambda^{-1/2}$. Then $\mathbf{I} + \mathbf{P}_\lambda\mathbf{E}_2 = \mathbf{G}_\lambda^{-1/2}(\mathbf{I}+\mathbf{M})\mathbf{G}_\lambda^{1/2}$, so $\mathbf{G}_\lambda^{1/2}(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_\lambda - \boldsymbol{\xi}) = -(\mathbf{I}+\mathbf{M})^{-1}\mathbf{M}\mathbf{G}_\lambda^{1/2}\boldsymbol{\xi}$, whose norm is at most $\frac{q}{1-q}\|\mathbf{G}_\lambda^{1/2}\boldsymbol{\xi}\|$. Finally $|\mathbf{e}_j^\top\mathbf{v}| \le \|\mathbf{G}_\lambda^{-1/2}\mathbf{e}_j\|\|\mathbf{G}_\lambda^{1/2}\mathbf{v}\|$. (iii) $\Pr[|\xi_j + R_j| > z s_j] \le \Pr[|\xi_j| > (z-\varpi)s_j] + \Pr[|R_j| > \varpi s_j]$. The first term is $2Q(z-\varpi) \le \alpha + 2\varpi\phi(z-\varpi)$. For the second, $q \le \|\mathbf{E}_2\|_{\mathrm{op}}/\lambda_{\min}(\mathbf{G}_\lambda)$ and Lemma 4.1(c) give $\Pr[q > \eta] \le 2e^{-p(\eta\vartheta_0-1)^2}$. On $\{q \le \eta\}$, (ii) and Markov's inequality with $\mathbb{E}\|\mathbf{G}_\lambda^{1/2}\boldsymbol{\xi}\|^2 = \operatorname{tr}(\mathbf{G}_\lambda\mathbf{V}_0) = \sigma_2^2\operatorname{tr}((\mathbf{I}+\mathbf{K})\mathbf{P}_\lambda)$ give the last term. The bound on ω_j uses $s_j^2 \ge \sigma_2^2\|\mathbf{P}_\lambda\mathbf{e}_j\|^2 \ge \sigma_2^2(\mathbf{P}_\lambda)_{jj}^2$ (as $\mathbf{K} \succeq 0$) and $\|\mathbf{K}\| \le 2\|\boldsymbol{\theta}\|^2$. (v) On Ω_L, $\vartheta_0 \ge \vartheta^\star$ (Lemma 5.1(b)); integrate (iii)–(iv) over $\mathcal{F}_1$ and add $\Pr[\Omega_L^c] \le \alpha_L$. □

The bound in (iii) is a certificate that the error vanishes as $\vartheta_0 \to \infty$, but it is loose (it uses the worst direction of $\mathbf{M}$ and Markov's inequality). The *typical* size of the remainder relative to s_j is governed by

$$\sigma_2 \operatorname{tr}(\mathbf{P}_\lambda) \;\le\; \frac{p\,\sigma_2}{\lambda_{\min}(\mathbf{G}_\lambda)} = \frac{\sqrt{p}}{2\vartheta_0},$$

the ratio of the $O(\sigma_2^2)$ mechanism bias $\sigma_2^2\mathbf{P}_\lambda\mathbf{M}_\chi(\mathbf{P}_\lambda)\boldsymbol{\theta}_\lambda$ to the $O(\sigma_2)$ standard deviation. At $\vartheta_0 = \vartheta^\star = 2.5$ and $p = 11$ this can reach 0.66 when all eigenvalues of $\mathbf{G}_\lambda$ sit at the floor. "First order in $1/\vartheta$" therefore hides a factor $\sqrt{p}$ (Correction 7). In the simulations (Table 4) $\sigma_2\operatorname{tr}(\mathbf{P}_\lambda) \le 0.18$ in the realistic scenarios and 0.59 in the stress scenario.

Theorem 7.2 (S5(b): bias-aware interval for β_j)**.** *Assume (M1) the working model on the clipped scale: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with (ε_i) conditionally independent given $\mathbf{X}$, $\mathbb{E}[\varepsilon_i \mid \mathbf{X}] = 0$ and $\operatorname{Var}(\varepsilon_i \mid \mathbf{X}) = s^2$; (M2) $\|\boldsymbol{\beta}\| \le B_\beta^{\mathrm{pub}}$ (possibly $+\infty$); (M3) $n > p$. Decompose*

$$\tilde{\theta}_j - \beta_j = b_j + S_j + \xi_j + R_j, \qquad b_j = -\lambda\,\mathbf{e}_j^\top\mathbf{P}_\lambda\Psi\boldsymbol{\beta}, \quad S_j = \mathbf{e}_j^\top\mathbf{P}_\lambda\mathbf{X}^\top\boldsymbol{\varepsilon}, \quad R_j = \tilde{\theta}_j - \theta_{\lambda,j} - \xi_j.$$

(i) [exact] $\mathbb{E}[\boldsymbol{\theta}_\lambda \mid \mathbf{X}, \mathbf{E}_1] = \boldsymbol{\beta} - \lambda\mathbf{P}_\lambda\Psi\boldsymbol{\beta}$; $|b_j| \le \lambda\|\Psi\mathbf{P}_\lambda\mathbf{e}_j\|\,\|\boldsymbol{\beta}\|$, with equality for $\boldsymbol{\beta} \parallel \Psi\mathbf{P}_\lambda\mathbf{e}_j$. $\operatorname{Var}(S_j \mid \mathbf{X}, \mathbf{E}_1) = s^2(\mathbf{P}_\lambda\mathbf{G}\mathbf{P}_\lambda)_{jj}$, and $\operatorname{Cov}(S_j, \xi_j) = 0$.

(ii) [exact] If $T \sim \mathcal{N}(b, v^2)$ and $\bar{B}$ is a random variable with $|b| \le \bar{B}$ on an event Ω, then $\Pr[|T| > v\,\mathrm{cv}_\alpha(\bar{B}/v)] \le \alpha + \Pr[\Omega^c]$, where $\mathrm{cv}_\alpha(t)$ is the $(1-\alpha)$-quantile of $|Z+t|$.

(iii) [exact on events] On $\Omega_L \cap \Omega_U$, $|b_j| \le \bar{b}_j^0 := \lambda\|\Psi\mathbf{P}_\lambda\mathbf{e}_j\|\hat{B}_\beta$.

(iv) [exact for the oracle interval] Let $v_j^2 = s^2(\mathbf{P}_\lambda\mathbf{G}\mathbf{P}_\lambda)_{jj} + (\mathbf{V}_0)_{jj}$. If $S_j + \xi_j \mid (\mathbf{X}, \mathbf{E}_1) \sim \mathcal{N}(0, v_j^2)$ and $R_j = 0$, then $\tilde{\theta}_j \pm v_j\,\mathrm{cv}_{\alpha_{\mathrm{CI}}}(\bar{b}_j^0/v_j)$ misses β_j with probability at most $\alpha_{\mathrm{CI}} + \alpha_L + \alpha_U'$, where $\alpha_U' = \alpha_U$ for the corrected U_{yy}^{c} and $2\alpha_U$ for the specification's U_{yy}.

(v) [first order] The reported interval replaces $\mathbf{P}_\lambda$ by $\tilde{\mathbf{P}}$ in $\bar{b}_j$ (relative error $\le \|\tilde{\mathbf{P}}\|\|\mathbf{E}_2\| = O_p(\vartheta_0^{-1})$, since $\mathbf{P}_\lambda - \tilde{\mathbf{P}} = \tilde{\mathbf{P}}\mathbf{E}_2\mathbf{P}_\lambda$), and v_j^2 by $V_{jj} = (\mathbf{V}_{\mathrm{mech}})_{jj} + \tilde{s}^2(\tilde{\mathbf{P}}\hat{\mathbf{G}}_2\tilde{\mathbf{P}})_{jj}$. It also drops R_j (Theorem 7.1(ii)) and ignores the dependence of $\mathbf{V}_0$ on $\boldsymbol{\theta}_\lambda$, hence on $\boldsymbol{\varepsilon}$, which is a relative $O(\|\boldsymbol{\theta}_\lambda - \mathbb{E}\boldsymbol{\theta}_\lambda\|/\|\boldsymbol{\theta}_\lambda\|)$ effect.

(vi) [asymptotic] Under (M1) the errors are bounded (clipping), so S_j is normal only asymptotically: by the Berry–Esseen bound for independent summands [11], $\sup_x|\Pr[S_j \le x\,\mathrm{sd}(S_j) \mid \mathbf{X}] - \Phi(x)| \le C_{\mathrm{BE}}\sum_i|a_{ij}|^3\mathbb{E}|\varepsilon_i|^3/(s^3\|\mathbf{a}_j\|^3)$ with $\mathbf{a}_j = \mathbf{X}\mathbf{P}_\lambda\mathbf{e}_j$. Together with (v): $\Pr[\beta_j \notin \mathrm{CI}_j,$ not refused$] \le \alpha_{\mathrm{CI}} + \alpha_L + \alpha_U' + \epsilon_j'$, with $\epsilon_j' \to 0$ when the Lyapunov ratio $\to 0$, $\vartheta^\star \to \infty$ and $B \to \infty$.

Proof. (i) $\mathbb{E}[\mathbf{c} \mid \mathbf{X}] = \mathbf{G}\boldsymbol{\beta}$ and $\mathbf{P}_\lambda\mathbf{G} = \mathbf{I} - \lambda\mathbf{P}_\lambda\Psi$; (λ, Ψ) is independent of $\boldsymbol{\varepsilon}$ given $\mathbf{X}$, because it depends on $\mathbf{G}$ and $\mathbf{E}_1$ only. Cauchy–Schwarz with $\Psi, \mathbf{P}_\lambda$ symmetric. $\mathbb{E}[\boldsymbol{\xi} \mid \mathbf{X}, \boldsymbol{\varepsilon}, \mathbf{E}_1] = 0$ gives the zero

covariance. (ii) $t \mapsto \Pr[|Z+t| \le c]$ is even and non-increasing in $|t|$ (Anderson's lemma [1]), so cv_α is non-decreasing in $|t|$. On Ω, $\mathrm{cv}_\alpha(\bar{B}/v) \ge \mathrm{cv}_\alpha(|b|/v)$, and $\Pr[|T| > v\,\mathrm{cv}_\alpha(|b|/v)] = \Pr[|Z + b/v| > \mathrm{cv}_\alpha(|b|/v)] = \alpha$. This is the construction of Armstrong and Kolesár [2], as used for regularized regression by Armstrong, Kolesár and Kwon [3]. (iii) Lemma 6.1(c) and (i). (iv) Apply (ii) with $\Omega = \Omega_L \cap \Omega_U$ and $\bar{B} = \bar{b}_j^0$. The dependence between $\bar{b}_j^0$ and T (through $\boldsymbol{\varepsilon}$ in U, and through $\mathbf{E}_1$) is harmless, because (ii) does not need $\bar{B}$ independent of T. (v), (vi) are as stated. □

Variance estimate. $\tilde{s}^2$ targets $\mathrm{RSS}_\lambda/(n-p)$, and $\mathbb{E}[\mathrm{RSS}_\lambda \mid \mathbf{X}] = s^2 \sum_i (1-h_i)^2 + \|\lambda \mathbf{X}\mathbf{P}_\lambda \Psi \boldsymbol{\beta}\|^2 \ge (n-p)s^2$, where $h_i \in [0,1)$ are the eigenvalues of $\mathbf{X}\mathbf{P}_\lambda\mathbf{X}^\top$ (at most p of them non-zero). So $\tilde{s}^2$ is conservative under the ridge (Table 5: 1.05, 1.48 and 1.79 against $s^2 = 1$ in three ridge scenarios). This conservativeness, and the worst-case nature of $\bar{b}_j$, explain the over-coverage of the β intervals.

Refusals and conditioning on the gate. All statements are unconditional, with a refusal producing no interval (equivalently, the trivial interval $\mathbb{R}$). Conditionally on passing, $\Pr[\text{miss} \mid \text{pass}] \le \Pr[\text{miss}, \text{pass}]/(1 - \Pr[\mathsf{refused}])$, with $\Pr[\mathsf{refused}]$ bounded in Lemma 5.1(d). The gate does not make the bootstrap moments finite (Lemma S6).

8 Lemma S6: the bootstrap RSS correction and covariance

Fix $\mathcal{F}_1$. Write $\mathrm{RSS}(\mathbf{G}', \mathbf{c}', yy') = yy' - \mathbf{c}'^\top\boldsymbol{\theta}' - \lambda\boldsymbol{\theta}'^\top\Psi\boldsymbol{\theta}'$ with $\boldsymbol{\theta}' = (\mathbf{G}' + \lambda\Psi)^{-1}\mathbf{c}'$; this equals $yy' - 2\mathbf{c}'^\top\boldsymbol{\theta}' + \boldsymbol{\theta}'^\top\mathbf{G}'\boldsymbol{\theta}'$, so $\mathrm{RSS}_\lambda := \mathrm{RSS}(\mathbf{G}, \mathbf{c}, \mathbf{y}^\top\mathbf{y}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}_\lambda\|^2$ and $\widetilde{\mathrm{RSS}} = \mathrm{RSS}(\hat{\mathbf{G}}_2, \hat{\mathbf{c}}_2, \widehat{yy})$. Define the normalized quantities

$$\mathbf{M} = \mathbf{G}_\lambda^{-1/2}\mathbf{E}_2\mathbf{G}_\lambda^{-1/2}, \quad \mathbf{h} = \mathbf{G}_\lambda^{-1/2}\mathbf{f}_2, \quad \boldsymbol{\phi} = \mathbf{G}_\lambda^{1/2}\boldsymbol{\theta}_\lambda, \quad \mathbf{N} = \lambda\mathbf{G}_\lambda^{-1/2}\Psi\mathbf{G}_\lambda^{-1/2} \ (0 \preceq \mathbf{N} \preceq \mathbf{I}),$$

so that, exactly, $\widetilde{\mathrm{RSS}} = \mathbf{y}^\top\mathbf{y} + g - (\boldsymbol{\phi}+\mathbf{h})^\top(\mathbf{I}+\mathbf{M})^{-1}\big[\mathbf{I} + \mathbf{N}(\mathbf{I}+\mathbf{M})^{-1}\big](\boldsymbol{\phi}+\mathbf{h})$. For a symmetric truncation level $\eta \in (0,1)$ let $T_\eta = \{\|\mathbf{M}\|_{\mathrm{op}} \le \eta\}$. In the *truncated bootstrap* a draw is kept iff $\|(\hat{\mathbf{G}}_2 + \lambda\Psi)^{-1/2}\mathbf{E}^*(\hat{\mathbf{G}}_2 + \lambda\Psi)^{-1/2}\|_{\mathrm{op}} \le \eta$; this event is invariant under $\mathbf{E}^* \mapsto -\mathbf{E}^*$.

Lemma 8.1 (S6)**.** *Under Assumption 2.1, conditionally on $\mathcal{F}_1$:*

(a) ***Expansion.*** $\widetilde{\mathrm{RSS}} = \mathrm{RSS}_\lambda + R_1 + R_2 + R_3 + \dots$, *converging on T_η, where R_k is homogeneous of degree k in $(\mathbf{M}, \mathbf{h}, g)$:*

$$R_1 = g - 2\boldsymbol{\theta}_\lambda^\top\mathbf{f}_2 + \boldsymbol{\theta}_\lambda^\top\mathbf{E}_2\boldsymbol{\theta}_\lambda - 2\lambda\boldsymbol{\theta}_\lambda^\top\Psi\boldsymbol{\xi}, \qquad R_2 = -\boldsymbol{\xi}^\top(\mathbf{G} + 2\lambda\Psi)\boldsymbol{\xi} + 2\lambda\boldsymbol{\theta}_\lambda^\top\Psi\mathbf{P}_\lambda\mathbf{E}_2\boldsymbol{\xi},$$

with $\boldsymbol{\xi} = \mathbf{P}_\lambda(\mathbf{f}_2 - \mathbf{E}_2\boldsymbol{\theta}_\lambda)$, and $|R_k| \le c_k\|\mathbf{M}\|^{k-2}\big(\|\mathbf{M}\boldsymbol{\phi}\| + \|\mathbf{h}\| + \|\mathbf{N}\boldsymbol{\phi}\|\big)\big(\|\mathbf{M}\boldsymbol{\phi}\| + \|\mathbf{h}\|\big)$ for $k \ge 2$, with combinatorial constants $c_k \le 4(k+1)$.

(b) ***Second-order bias.*** $\mathbb{E}[R_1] = \mathbb{E}[R_3 \mid T_\eta] = 0$ *(odd degree, symmetric event), and*

$$\mathbb{E}[R_2] = b_{\mathrm{RSS}}(\mathbf{G}, \boldsymbol{\theta}_\lambda) := -\sigma_2^2\Big[\mathrm{tr}\big((\mathbf{G} + 2\lambda\Psi)\mathbf{P}_\lambda(\mathbf{I} + \mathbf{K}(\boldsymbol{\theta}_\lambda))\mathbf{P}_\lambda\big) + 2\lambda\,\boldsymbol{\theta}_\lambda^\top\Psi\mathbf{P}_\lambda\mathbf{M}_\chi(\mathbf{P}_\lambda)\boldsymbol{\theta}_\lambda\Big].$$

Hence $\mathbb{E}[\widetilde{\mathrm{RSS}} \mid T_\eta] - \mathrm{RSS}_\lambda = b_{\mathrm{RSS}}(\mathbf{G}, \boldsymbol{\theta}_\lambda) + \rho_1$, with $|\rho_1| \le C_\eta\,\vartheta_0^{-2}\,\mathcal{S}_2 + O(e^{-p(\eta\vartheta_0 - 1)^2})$, where $\mathcal{S}_2$ is the scale of the second-order terms, $\mathcal{S}_2 \asymp \sigma_2^2(1 + \|\boldsymbol{\theta}_\lambda\|^2)\,\mathrm{tr}(\mathbf{P}_\lambda) + \|\mathbf{N}\boldsymbol{\phi}\|\,\sigma_2(1 + \|\boldsymbol{\theta}_\lambda\|)\sqrt{\mathrm{tr}\,\mathbf{P}_\lambda}/\vartheta_0$.

(c) ***RSS correction.*** *Let $\mathrm{Corr}_\infty = \mathbb{E}^*[\mathrm{RSS}^* \mid T_\eta^*] - \widetilde{\mathrm{RSS}}$ be the $B \to \infty$ limit of* Corr *for the truncated bootstrap. Then $\mathrm{Corr}_\infty = b_{\mathrm{RSS}}(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}}) + \hat{\rho}_1$, and*

$$\mathbb{E}\big[\widetilde{\mathrm{RSS}} - \mathrm{Corr}_\infty \;\big|\; T_\eta\big] - \mathrm{RSS}_\lambda = O\big(\vartheta_0^{-2}\mathcal{S}_2\big).$$

The uncorrected bias $b_{\mathrm{RSS}} = O(\mathcal{S}_2)$ is removed up to a remainder smaller by a factor ϑ_0^{-2}: $O(\sigma_2^2)$ becomes $O(\sigma_2^4)$. The Monte Carlo error of Corr *with B draws is $O_p(\mathcal{S}_1/\sqrt{B})$ with $\mathcal{S}_1^2 = \mathrm{Var}(R_1) = \sigma_{yy}^2 + \sigma_2^2\, O\big((1+\|\boldsymbol{\theta}_\lambda\|^2)^2\big)$. Antithetic pairs $(\mathbf{E}^*, \mathbf{f}^*, g^*), (-\mathbf{E}^*, -\mathbf{f}^*, -g^*)$ cancel R_1 and R_3 exactly and reduce it to $O_p(\mathcal{S}_2/\sqrt{B})$.*

(d) ***Covariance.*** *$\mathrm{Var}(\tilde{\boldsymbol{\theta}} \mid T_\eta) = \mathbf{V}_0\,(1+O(\vartheta_0^{-2}))$ and $\mathrm{Cov}^*(\boldsymbol{\theta}^* \mid T_\eta^*) = \mathbf{V}_0(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}})(1+O(\hat{\vartheta}^{-2}))$, with $\mathbf{V}_0(\mathbf{G}', \boldsymbol{\theta}') = \sigma_2^2(\mathbf{G}'+\lambda\Psi)^{-1}(\mathbf{I}+\mathbf{K}(\boldsymbol{\theta}'))(\mathbf{G}'+\lambda\Psi)^{-1}$; and $\mathbf{V}_0(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}}) = \mathbf{V}_0(\mathbf{G}, \boldsymbol{\theta}_\lambda)(1+O_p(\vartheta_0^{-1}))$, in the sense of the diagonal entries relative to themselves. So $\mathbf{V}_{\mathrm{mech}}$ estimates the mechanism covariance to first order, with relative error $O_p(\vartheta_0^{-1}) + O(\vartheta_0^{-2}) + O_p(B^{-1/2})$.*

(e) ***Untruncated bootstrap.*** *Without truncation, $\mathbb{E}^*\|\boldsymbol{\theta}^*\| = \infty$ (already for $p = 1$, $\mathbb{E}|1/(a+e)| = \infty$ for Gaussian e), so $\mathrm{Cov}_b(\boldsymbol{\theta}_b^*)$ and $\overline{\mathrm{RSS}^*}$ do not converge as $B \to \infty$. For fixed B they coincide with the truncated versions except on an event of probability $\le B \cdot 2e^{-p(\eta\hat{\vartheta}-1)^2}$; the gate $\hat{\vartheta} \ge 1$ alone does not make this small.*

Proof [first order: constants c_k, C_η and the scale $\mathcal{S}_2$ are not made fully explicit]. (a) Expand $(\mathbf{I}+\mathbf{M})^{-1} = \sum_k (-\mathbf{M})^k$, which converges on T_η. Every term of degree $k \ge 2$ in $(\mathbf{M}, \mathbf{h})$ is a bilinear form whose two outer vectors are $\mathbf{h}$, $\mathbf{M}\boldsymbol{\phi}$ or $\mathbf{N}\boldsymbol{\phi}$, with at most one of them $\mathbf{N}\boldsymbol{\phi}$ and at most $k-2$ further factors $\mathbf{M}$ or $\mathbf{N}$ between them ($\|\mathbf{N}\| \le 1$); the number of such terms is at most $4(k+1)$ (outer pairs $(\boldsymbol{\phi}, \boldsymbol{\phi})$: $k+2$ terms; $(\mathbf{h}, \boldsymbol{\phi})$ and $(\boldsymbol{\phi}, \mathbf{h})$: $k+1$ each; $(\mathbf{h}, \mathbf{h})$: k), each with coefficient ± 1. The explicit R_1, R_2 follow by differentiation: with $\boldsymbol{\theta}' = \boldsymbol{\theta}_\lambda + \boldsymbol{\xi} + \boldsymbol{\zeta} + \ldots$, $\boldsymbol{\zeta} = -\mathbf{P}_\lambda \mathbf{E}_2 \boldsymbol{\xi}$, use $\mathbf{c} - \mathbf{G}\boldsymbol{\theta}_\lambda = \lambda\Psi\boldsymbol{\theta}_\lambda$ and $\mathbf{f}_2 - \mathbf{E}_2\boldsymbol{\theta}_\lambda = \mathbf{G}_\lambda\boldsymbol{\xi}$.

(b) Odd-degree terms have zero mean under the symmetric Gaussian law and on the symmetric event T_η. $\mathbb{E}[\boldsymbol{\xi}^\top \mathbf{A}\boldsymbol{\xi}] = \mathrm{tr}(\mathbf{A}\mathbf{V}_0)$ and $\mathbb{E}[\mathbf{E}_2\mathbf{P}_\lambda\mathbf{E}_2] = \sigma_2^2\mathbf{M}_\chi(\mathbf{P}_\lambda)$ give b_{RSS}; $\mathbb{E}[\mathbf{E}_2\mathbf{P}_\lambda\mathbf{f}_2] = 0$. The degree-4 terms are bounded via (a) by $c_4\mathbb{E}[\|\mathbf{M}\|^2(\cdot)(\cdot)]$, and $\mathbb{E}\|\mathbf{M}\|^2 \le (\mathbb{E}\|\mathbf{E}_2\|^2)/\lambda_{\min}(\mathbf{G}_\lambda)^2 = \vartheta_0^{-2}(1 + O(p^{-1/2}))$ by Lemma 4.1(d) and (b); also $\mathbb{E}\|\mathbf{h}\|^2 = \sigma_2^2\,\mathrm{tr}\,\mathbf{P}_\lambda$ and $\mathbb{E}\|\mathbf{M}\boldsymbol{\phi}\|^2 = \sigma_2^2\,\mathrm{tr}(\mathbf{P}_\lambda\mathbf{K}(\boldsymbol{\theta}_\lambda)) \le 2\sigma_2^2\|\boldsymbol{\theta}_\lambda\|^2\,\mathrm{tr}\,\mathbf{P}_\lambda$. Degrees ≥ 5 are summed geometrically on T_η. Removing the conditioning costs $\Pr[T_\eta^c] \le 2e^{-p(\eta\vartheta_0-1)^2}$ (Lemma 4.1(c)) times bounded conditional moments.

(c) Apply (b) in the bootstrap world centred at $(\hat{\mathbf{G}}_2, \hat{\mathbf{c}}_2, \widehat{yy})$, where the bootstrap law is exactly the mechanism law. Then $b_{\mathrm{RSS}}(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}}) - b_{\mathrm{RSS}}(\mathbf{G}, \boldsymbol{\theta}_\lambda)$ is a smooth function of $(\mathbf{E}_2, \mathbf{f}_2)$ whose linear part has zero mean; its quadratic part is $O(\vartheta_0^{-2}\mathcal{S}_2)$ in mean, since b_{RSS} is homogeneous of degree 2 in the noise scale and every derivative brings a factor of order ϑ_0^{-1}. The Monte Carlo statements follow from $\mathrm{Var}(\mathrm{RSS}^*) = \mathrm{Var}(R_1^*) + O(\mathcal{S}_2^2)$, and the antithetic average is even in the noise.

(d) $\mathrm{Var}(\tilde{\boldsymbol{\theta}}) = \mathrm{Var}(\boldsymbol{\xi}) + 2\,\mathrm{Cov}(\boldsymbol{\xi}, \boldsymbol{\zeta}) + \ldots$; $\mathrm{Cov}(\boldsymbol{\xi}, \boldsymbol{\zeta})$ is an odd (third) moment and vanishes, and the remaining terms are $O(\vartheta_0^{-2})$ relative by (a). The plug-in error comes from $\tilde{\mathbf{P}} - \mathbf{P}_\lambda = -\tilde{\mathbf{P}}\mathbf{E}_2\mathbf{P}_\lambda$ ($\|\cdot\| \le \|\tilde{\mathbf{P}}\|\|\mathbf{E}_2\|\|\mathbf{P}_\lambda\|$, relative $O_p(\vartheta_0^{-1})$) and $\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_\lambda = O_p(\|\boldsymbol{\xi}\|)$.

(e) For $p = 1$, $\int |a+e|^{-1}\phi_\sigma(e)\,de = \infty$ because the density is positive at $e = -a$. For general p, $\|(\hat{\mathbf{G}}_2 + \mathbf{E}^* + \lambda\Psi)^{-1}\| \ge 1/|\lambda_{\min}(\cdot)|$, and the law of $\lambda_{\min}$ has a positive density at 0. Moreover $\hat{\vartheta} \ge 1$ only gives $\lambda_{\min}(\hat{\mathbf{G}}_2 + \lambda\Psi) \ge 2\sigma_2\sqrt{p}$, which for $p = 11$ is only $1.33\,\mathbb{E}\lambda_{\max}(\mathbf{E}^*)$, so near-singular draws are not rare when $\hat{\vartheta}$ is close to 1. □

Closed-form alternative. Because b_{RSS} and $\mathbf{V}_0$ are explicit, R may replace the Monte Carlo by $\mathrm{Corr}^{\mathrm{cf}} = b_{\mathrm{RSS}}(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}})$ and $\mathbf{V}_{\mathrm{mech}}^{\mathrm{cf}} = \mathbf{V}_0(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}})$. These have the same first-order guarantees as the truncated bootstrap, no Monte Carlo error, and no moment problem. They are post-processing, so they cost no privacy.

Numerical checks (`s3_s6_sim.py`, fixed design and data, 2×10^4 stage-2 releases for the "truth", 200 independent stage-2 releases each with $B = 400$ bootstrap draws). Table 4 shows that b_{RSS} matches the Monte Carlo bias within 2% when $\sigma_2\,\mathrm{tr}\,\mathbf{P}_\lambda \le 0.18$ and within 5% at 0.59. $\mathbf{V}_0$ and

$\mathbf{V}_{\text{mech}}$ match the true mechanism standard deviations within 2% in the realistic scenarios. The spread of the bootstrap standard errors across releases (3.5–4%) is essentially the Monte Carlo error $1/\sqrt{2B} = 3.5\%$. Antithetic pairs reduce the Monte Carlo s.d. of Corr by a factor of 5–100 at the same cost. In the stress scenario ($\vartheta_0 = 2.52$, outside the default design) $\mathbf{V}_0$ is 2–6% too small, the bootstrap is 3–10% too large, and Corr overshoots the bias by 12%. The corrected RSS still removes 87% of the bias.

scenario	ϑ_0	$\sigma_2 \operatorname{tr} \mathbf{P}_\lambda$	$\mathbb{E}[\widetilde{\mathrm{RSS}}] - \mathrm{RSS}_\lambda$ (MC)	b_{RSS}	Corr mean (sd)	antithetic (sd)	$\max_j \lvert \mathrm{sd}_{\mathbf{V}_{\text{mech}}}/\mathrm{sd} - 1 \rvert$
off ($\lambda = 0$)	48.7	0.017	-0.106 ± 0.062	-0.107	-0.08 (0.44)	-0.107 (0.004)	0.009
ridge $\mathbf{I}$	7.07	0.145	-5.19 ± 0.27	-5.29	-5.42 (1.90)	-5.33 (0.21)	0.017
ridge Π_O	8.05	0.111	-2.92 ± 0.13	-2.90	-3.03 (0.99)	-2.94 (0.12)	0.015
small n	8.17	0.183	-6.97 ± 0.16	-6.96	-6.98 (1.14)	-7.04 (0.23)	0.018
stress	2.52	0.588	-23.61 ± 0.17	-22.49	-26.48 (1.98)	-26.71 (1.58)	0.097

Table 4: Lemma S6 at fixed data and fixed (λ, Ψ). "±" is the Monte Carlo standard error; "sd" is the s.d. across 200 independent releases.

9 S7: the exact stage-1 radius and the transfer to the protocol

The exact radius of settings I and P replaces the union bound of Lemma S2(c) by a Monte Carlo quantile of $\lambda_{\max}$ of the unit stage-1 noise matrix. Let F_{d_A,d_O} be the distribution function of $\lambda_{\max}(\mathbf{E})$ for a masked symmetric noise matrix of unit scale with blocks (d_A, d_O), and $q_{1-\alpha} = F^{-1}_{d_A,d_O}(1-\alpha)$. Draw M independent copies, sort their largest eigenvalues $L_{(1)} \le \cdots \le L_{(M)}$, let j be the smallest integer with $\Pr[\mathrm{Bin}(M, 1-\alpha) \le j] \ge c$, and set $\hat{q} = L_{(j+1)}$, $r_1^{\mathrm{ex}} = \sigma_1 \hat{q}$ (the implementation's rule, with $c = 0.99$ and $M = \max\{2 \cdot 10^5, \lceil 2500/\alpha \rceil\}$). For release s let

$$\bar{w}_s = \frac{N}{2\sqrt{2}\,\Omega_f} + \sqrt{2\pi}\, 2^{-\kappa} \sigma_s + \bar{d}_s, \qquad \pi_s = \delta_B + N \,\mathrm{TV}_1(\sigma_{c,s}) + 2Ne^{-t_\eta^2/2},$$

with $\bar{d}_s$ the proved bound on the decoding offset (crypto proofs, Part A, Proposition on the decoding offset; it includes the rounding of the centre) and TV_1 bounded by Lemma S0(b).

Proposition 9.1 (S7: exact stage-1 radius)**.** *(a) The law of $\lambda_{\max}(\mathbf{E}_1)/\sigma_1$ is F_{d_A,d_O}, which depends only on (d_A, d_O) and is continuous.*

(b) Over the Monte Carlo draws, $\Pr[\hat{q} \ge q_{1-\alpha}] \ge c$.

(c) If the draws are independent of the release, $\Pr[\lambda_{\max}(\mathbf{E}_1) > r_1^{\mathrm{ex}}] = (M-j)/(M+1)$, at most α whenever $j + 1 \ge (1-\alpha)(M+1)$.

(d) Under the protocol, (c) holds with r_1^{ex} enlarged by $\sqrt{2}\,\bar{w}_1$ and the probability enlarged by π_1; so do Lemma S2(c) and the radius event of Lemma S3.

Proof. (a) $\mathbf{E}_1 = \sigma_1 \mathbf{E}$ with $\mathbf{E}$ of unit scale, whose law is fixed by the mask. $\lambda_{\max}(\mathbf{E}) \ge 0$ by interlacing (the R block is zero). If $\mathbf{C} \ne 0$, pick $\mathbf{a}, \mathbf{v}$ with $\mathbf{a}^\top \mathbf{C} \mathbf{v} > 0$; the vector $(\mathbf{a}, t\mathbf{v})$ gives $2t\mathbf{a}^\top \mathbf{C}\mathbf{v} + t^2 \mathbf{v}^\top \mathbf{B}\mathbf{v} > 0$ for small $t > 0$; so $\Pr[\lambda_{\max}(\mathbf{E}) = 0] \le \Pr[\mathbf{C} = 0] = 0$. For $t > 0$, $\{\lambda_{\max} \le t\}$ is a convex set of the Gaussian coordinates with non-empty interior (it contains a neighbourhood of $\mathbf{C} = 0$, $\mathbf{B} = -\mathbf{I}$), and $\{\lambda_{\max} = t\}$ lies in its boundary, which has Lebesgue measure zero. (b) $F(q_{1-\alpha}) = 1-\alpha$ by continuity, the number K of draws below $q_{1-\alpha}$ is $\mathrm{Bin}(M, 1-\alpha)$, and $\hat{q} < q_{1-\alpha}$ iff $K \ge j+1$, of probability $1 - \Pr[K \le j] \le 1 - c$. (c) $F(L_i)$ are independent uniforms, so $F(\hat{q})$ is $\mathrm{Beta}(j+1, M-j)$ with mean $(j+1)/(M+1)$; conditionally on the draws the exceedance probability is $1 - F(\hat{q})$. (d) Lemma S0(b),(c) couple each release with Gaussian slot noise; on the coupling event

(probability $\geq 1-\pi_s$: coupling, certificate, no wraparound) the decoded entries differ from $\mathbf{z}_s + \boldsymbol{\xi}_s^G$ by a vector of norm at most $\bar{w}_s$ (the rounding of $\mathbf{Z}$, the flooded error, and the decoding offset), and the symmetric matrix assembled from it has operator norm at most $\sqrt{2}\,\bar{w}_s$; Weyl's inequality gives the enlargement. □

At $p = 11$, $d_A = 5$, $\alpha = 0.0025$: $M = 10^6$, $j = 997{,}615$, $\hat{q} = 7.415$ against 11.529 for the union bound, and the exact level is $(M-j)/(M+1) = 0.002385$. At the evaluation setting $N\,\mathrm{TV}_1(\sigma_{c,s}) \leq 2^{-122.2}$ ($\approx 10^{-37}$) and $\bar{w}_s$ is between $2\cdot 10^{-8}$ ($\bar{n}_O = 4{,}096$) and $5\cdot 10^{-6}$ ($\bar{n}_O = 10^6$, every row matchable), against $\sigma_1 \geq 128$. With the fixed public seed used in practice $\hat{q}$ is one number; (b) says that the procedure producing it errs with probability at most $1-c$, and (c) is a statement about the procedure with fresh draws. Parts (a)–(c) and the structure of (d) follow the companion book (Chapter 13, Proposition `inf.r1exact`); the book's caveat on (d), a numerically checked total-variation constant, is removed by Lemma S0(b) below, and its offset bound is proved in Part A.

10 S8: where the ridge switches off

Write $g_1^\star = g^\star + r_1$. The stage-1 rule returns $\lambda = 0$ iff $\lambda_{\min}(\hat{\mathbf{G}}_1) \geq g_1^\star$: then $\lambda_I = \lambda_O = 0$; conversely $\lambda_I > 0$ forces $\lambda_O > 0$, since $\lambda_O = 0$ means $\lambda_{\min}(\hat{\mathbf{G}}_1) \geq g_1^\star$.

Proposition 10.1 (S8: a bracket for the switch-off)**.** *Condition on the data, let $\alpha' \in (0,1)$, and let $q'_{1-\alpha'}$ be the $(1-\alpha')$ quantile of $\lambda_{\max}$ of the unit stage-1 noise matrix (or its union bound $2\sqrt{p} + 2\sqrt{\ln(1/\alpha')}$). Under Assumption 2.1:*

(a) if $\lambda_{\min}(\mathbf{G}) \geq g_1^\star + \sigma_1 q'_{1-\alpha'}$, the rule returns $\lambda = 0$ with probability at least $1-\alpha'$;

(b) if $\lambda_{\min}(\mathbf{G}) \leq g_1^\star - \sqrt{2}\,\sigma_1 z_{1-\alpha'}$, it returns $\lambda > 0$ with probability at least $1-\alpha'$;

(c) the threshold in (a) cannot be lowered in terms of $\lambda_{\min}(\mathbf{G})$ alone: for $\mathbf{G} = \lambda_{\min}(\mathbf{G})\mathbf{I}$ the rule returns $\lambda = 0$ with probability exactly $F_{d_A,d_O}((\lambda_{\min}(\mathbf{G}) - g_1^\star)/\sigma_1)$.

Under the protocol the thresholds move by $\sqrt{2}\,\bar{w}_1$ and the probabilities by π_1 (Proposition 9.1(d)).

Proof. (a) Weyl: $\lambda_{\min}(\hat{\mathbf{G}}_1) \geq \lambda_{\min}(\mathbf{G}) - \lambda_{\max}(-\mathbf{E}_1)$, and $\lambda_{\max}(-\mathbf{E}_1)$ has the law of $\sigma_1\lambda_{\max}(\mathbf{E})$. (b) For a unit bottom eigenvector $\mathbf{u}$ of $\mathbf{G}$, $\lambda_{\min}(\hat{\mathbf{G}}_1) \leq \lambda_{\min}(\mathbf{G}) + \mathbf{u}^\top\mathbf{E}_1\mathbf{u}$, a centred Gaussian with variance $\sigma_1^2(\sum_{i\in O} u_i^4 + 4\sum_{i<j,\chi_{ij}=1} u_i^2u_j^2) \leq 2\sigma_1^2$. (c) $\lambda_{\min}(c\mathbf{I} + \mathbf{E}_1) = c - \lambda_{\max}(-\mathbf{E}_1)$. □

With $\lambda_{\min}(\mathbf{G}) \approx n\ell$ the switch-off lies between $(1 - \sqrt{2}\sigma_1 z_{1-\alpha'}/g_1^\star)n_{\mathrm{off}}$ and $(1 + \sigma_1 q'_{1-\alpha'}/g_1^\star)n_{\mathrm{off}}$; on the simulation population at $\alpha' = 0.05$ ($q'_{0.95} = 6.303$, $\sigma_2/\sigma_1 = 0.645$) this is $[0.87, 1.35]\,n_{\mathrm{off}}$ under setting I and $[0.79, 1.58]\,n_{\mathrm{off}}$ under setting P, at every ε.

Lemma 10.2 (S8′: the size of the downward bias of $\lambda_{\min}$)**.** *For fixed symmetric $\mathbf{G}$ and a masked symmetric noise matrix $\mathbf{E}$ of scale σ (any variances $\leq \sigma^2$ off the diagonal and $\leq 2\sigma^2$ on it), with $\mathbf{g} \sim \mathcal{N}(0, \mathbf{I}_p)$:*

(a) $\mathbb{E}\lambda_{\min}(\mathbf{G}+\mathbf{E}) \leq \lambda_{\min}(\mathbf{G})$;

(b) $\mathbb{E}\lambda_{\min}(\mathbf{G}+\mathbf{E}) \geq m(\mathbf{G},\sigma) := \mathbb{E}\min_{\|\mathbf{u}\|=1}\big(\mathbf{u}^\top\mathbf{G}\mathbf{u} - 2\sigma\mathbf{g}^\top\mathbf{u}\big) \geq \lambda_{\min}(\mathbf{G}) - 2\sigma\mathbb{E}\|\mathbf{g}\|$; $m(\mathbf{G},\sigma)$ depends only on the eigenvalues $\mu_1 \leq \cdots \leq \mu_p$ of $\mathbf{G}$ and equals $\mathbb{E}[\theta - \sum_k \sigma^2 g_k^2/(\mu_k - \theta)]$, with $\theta < \mu_1$ the root of $\sum_k \sigma^2 g_k^2/(\mu_k-\theta)^2 = 1$;

(c) $\Pr[\lambda_{\min}(\mathbf{G}+\mathbf{E}) \leq \mathbb{E}\lambda_{\min}(\mathbf{G}+\mathbf{E}) - t] \leq e^{-t^2/(4\sigma^2)}$ for $t \geq 0$.

Proof. (a) $\lambda_{\min}(\mathbf{G}+\mathbf{E}) \leq \mathbf{u}^\top\mathbf{G}\mathbf{u} + \mathbf{u}^\top\mathbf{E}\mathbf{u}$ for a bottom eigenvector $\mathbf{u}$ of $\mathbf{G}$, and $\mathbb{E}\mathbf{u}^\top\mathbf{E}\mathbf{u} = 0$. (b) $-\lambda_{\min}(\mathbf{G}+\mathbf{E}) = \sup_{\mathbf{u}}(-\mathbf{u}^\top\mathbf{G}\mathbf{u} - X_{\mathbf{u}})$ with $X_{\mathbf{u}} = \mathbf{u}^\top\mathbf{E}\mathbf{u}$. The processes $(-\mathbf{u}^\top\mathbf{G}\mathbf{u} - X_{\mathbf{u}})$ and $(-\mathbf{u}^\top\mathbf{G}\mathbf{u} + Y_{\mathbf{u}})$, $Y_{\mathbf{u}} = 2\sigma\mathbf{g}^\top\mathbf{u}$, are Gaussian with equal means, and by the proof of Lemma S2(a)

$\mathbb{E}(X_{\mathbf{u}} - X_{\mathbf{v}})^2 \le \mathbb{E}(Y_{\mathbf{u}} - Y_{\mathbf{v}})^2$. The Sudakov–Fernique inequality holds for Gaussian vectors with equal means (its interpolation proof uses only the covariances; Chatterjee [7]), and it extends to the sphere by continuity; so $\mathbb{E}[-\lambda_{\min}(\mathbf{G}+\mathbf{E})] \le \mathbb{E}\sup_{\mathbf{u}}(-\mathbf{u}^\top\mathbf{G}\mathbf{u} + 2\sigma\mathbf{g}^\top\mathbf{u})$, which is (b). The minimum over the sphere of $\mathbf{u}^\top\mathbf{G}\mathbf{u} - 2\mathbf{h}^\top\mathbf{u}$ is a trust-region problem whose value is $\theta - \mathbf{h}^\top(\mathbf{G}-\theta\mathbf{I})^{-1}\mathbf{h}$ at the root $\theta < \mu_1$ of $\|(\mathbf{G}-\theta\mathbf{I})^{-1}\mathbf{h}\| = 1$ (the hard case has probability zero), and $\mathbf{g}$ is rotation invariant. The last inequality takes $\mathbf{u}^\top\mathbf{G}\mathbf{u} \ge \lambda_{\min}(\mathbf{G})$. (c) $\lambda_{\min}(\mathbf{G}+\mathbf{E})$ is an infimum of affine functions of the Gaussian coordinates with Lipschitz constants at most $\sqrt{2}\sigma$ (proof of Lemma S2(b)); Gaussian concentration. □

Remark 10.3 (What the lemma gives, and what remains open)**.** On the simulation population (Σ of the clipped, standardized design, eigenvalues $0.599, 0.621, 0.680, 0.754, \ldots$, $p = 11$), at $n = n_{\text{off}}$ under setting I and $\varepsilon = 1$, (b) bounds the downward bias by $3.19\,\sigma_1$, against $1.70\,\sigma_1$ in simulation, and $2\sigma_1\mathbb{E}\|\mathbf{g}\| \approx 6.5\,\sigma_1$ for $\mathbf{G} \propto \mathbf{I}$ (script `experiments/proofs_check/lmin_bias_bound.py`). Combined with (c), it guarantees the switch-off with probability 0.95 from $1.4\,n_{\text{off}}$ under setting I (above $1.5\,n_{\text{off}}$ under P), which is weaker than the bracket's 1.35 (and 1.58). By Proposition 10.1(c) no bound in terms of $\lambda_{\min}(\mathbf{G})$ alone can do better than the bracket; the empirical factors 1.2 (I) and 1.4 (P) reflect the spread of the population's spectrum. Turning them into bounds needs a tail bound for $\lambda_{\min}(\mathbf{G}+\mathbf{E})$ that uses the spectrum and is sharp to within much less than the factor 1.9 that the comparison of (b) loses: second-order perturbation, $\lambda_{\min}(\mathbf{G}+\mathbf{E}) \approx \mu_1 + \mathbf{u}_1^\top\mathbf{E}\mathbf{u}_1 - \sum_{k\ge2}(\mathbf{u}_k^\top\mathbf{E}\mathbf{u}_1)^2/(\mu_k - \mu_1)$, needs gaps $\mu_k - \mu_1$ above about $2\|\mathbf{E}\|$, and at n_{off} the three smallest eigenvalues of $n\Sigma$ lie within $0.1\,n\ell \approx 2.4\,\sigma_1$ of each other. This remains [open].

11 S9: mechanism bias with explicit constants

Fix (λ, Ψ) independently of the stage-2 noise, write $\mathbf{G}$ for $\mathbf{G}_\lambda$, $\mathbf{P} = \mathbf{G}^{-1}$, $\sigma = \sigma_2$, $\vartheta = \lambda_{\min}(\mathbf{G})/(2\sigma\sqrt{p})$, $\mathbf{H} = \mathbf{P}\mathbf{E}$, $\mathbf{M} = \mathbf{G}^{-1/2}\mathbf{E}\mathbf{G}^{-1/2}$, $q = \|\mathbf{M}\|_{\text{op}}$, and for $\bar\eta \in (0,1)$ let $T = \{q \le \bar\eta\}$; T is invariant under $\mathbf{E} \mapsto -\mathbf{E}$. Let $\kappa_p = (2\sqrt{p} + c_0 + 2\sqrt{2})/(2\sqrt{p})$ ($\kappa_p = 1.81$ at $p = 11$), so that $(\mathbb{E}\|\mathbf{E}\|_{\text{op}}^4)^{1/4} \le 2\sigma\sqrt{p}\,\kappa_p$ (Lemma S2(d)) and $\mathbb{E}q^4 \le \kappa_p^4\vartheta^{-4}$. The bias matrix of the paper is $\mathbf{P}\mathbf{M}_\chi(\mathbf{P})$, with $\mathbf{M}_\chi$ the contraction of Section 2.

Theorem 11.1 (S9: leading-order conditional bias, explicit remainder)**.** *If $\bar\eta\vartheta > 1$, then $\mathbb{E}[\tilde{\boldsymbol\theta} \mid T]$ exists and $\mathbb{E}[\tilde{\boldsymbol\theta} \mid T] - \boldsymbol\theta_\lambda = \sigma^2\mathbf{P}\mathbf{M}_\chi(\mathbf{P})\boldsymbol\theta_\lambda + \boldsymbol\rho$ with*

$$\|\boldsymbol\rho\|_{\mathbf{G}} \le \frac{1}{\Pr(T)}\Big[\frac{\kappa_p^4}{(1-\bar\eta)\vartheta^4}\big(\|\boldsymbol\theta_\lambda\|_{\mathbf{G}} + \sigma\sqrt{\operatorname{tr}\mathbf{P}}\big) + \frac{2\sqrt{2}\,\kappa_p^2}{\vartheta^2}e^{-p(\bar\eta\vartheta-1)^2/2}\|\boldsymbol\theta_\lambda\|_{\mathbf{G}}\Big],$$

where $\Pr(T^c) \le 2e^{-p(\bar\eta\vartheta-1)^2}$, and $\|\sigma^2\mathbf{P}\mathbf{M}_\chi(\mathbf{P})\boldsymbol\theta_\lambda\|_{\mathbf{G}} \le \kappa_p^2\vartheta^{-2}\|\boldsymbol\theta_\lambda\|_{\mathbf{G}}$. The moment noise $\mathbf{f}$ contributes nothing at order σ^2. For $\Psi = \mathbf{I}$, $\|\sigma^2\mathbf{P}\mathbf{M}_\chi(\mathbf{P})\boldsymbol\theta_\lambda\| \le (3 + p + d_O)\sigma^2\|\mathbf{c}\|/(\lambda_{\min}(\mathbf{G}) + \lambda)^3$ with $\mathbf{G}$ the unpenalized Gram; for $\Psi = \Pi_O$ and $\mathbf{A}$ invertible it is $O(\sigma^2/\lambda)$.

Proof. Expansion. On T, $(\mathbf{I}+\mathbf{H})^{-1} = \mathbf{I} - \mathbf{H} + \mathbf{H}^2 - \mathbf{H}^3 + \mathbf{H}^4(\mathbf{I}+\mathbf{H})^{-1}$, so $\tilde{\boldsymbol\theta} = (\mathbf{I} - \mathbf{H} + \mathbf{H}^2 - \mathbf{H}^3)(\boldsymbol\theta_\lambda + \mathbf{P}\mathbf{f}) + \mathbf{R}_4$ with $\|\mathbf{R}_4\|_{\mathbf{G}} \le \frac{q^4}{1-q}(\|\boldsymbol\theta_\lambda\|_{\mathbf{G}} + \|\mathbf{G}^{-1/2}\mathbf{f}\|)$, because $\mathbf{G}^{1/2}\mathbf{H}^k\mathbf{G}^{-1/2} = \mathbf{M}^k$; and $\|\tilde{\boldsymbol\theta}\|_{\mathbf{G}} \le (1-\bar\eta)^{-1}\|\mathbf{G}^{-1/2}(\mathbf{c}+\mathbf{f})\|$ on T, which is integrable. *Odd terms.* $\mathbb{E}[\mathbf{H}^m \mid T] = 0$ for odd m by symmetry, and every term containing $\mathbf{P}\mathbf{f}$ has zero conditional mean ($\mathbf{f}$ is centred and independent of $\mathbf{E}$). *Remainder.* $\mathbb{E}[q^4\mathbf{1}_T] \le \kappa_p^4\vartheta^{-4}$, and $\mathbb{E}\|\mathbf{G}^{-1/2}\mathbf{f}\| \le \sigma\sqrt{\operatorname{tr}\mathbf{P}}$, independently of q. *Conditioning.* $\mathbb{E}[\mathbf{H}^2 \mid T] - \mathbb{E}\mathbf{H}^2 = (\mathbb{E}[\mathbf{H}^2]\Pr(T^c) - \mathbb{E}[\mathbf{H}^2\mathbf{1}_{T^c}])/\Pr(T)$, of $\mathbf{G}$-operator norm at most $2(\mathbb{E}q^4)^{1/2}\Pr(T^c)^{1/2}/\Pr(T)$ by Cauchy–Schwarz; $\Pr(T^c) \le \Pr[\|\mathbf{E}\|_{\text{op}} > 2\bar\eta\vartheta\sigma\sqrt{p}] \le 2e^{-p(\bar\eta\vartheta-1)^2}$ by Lemma S2(c). *Contraction.* $\mathbb{E}\mathbf{H}^2 = \mathbf{P}\mathbb{E}[\mathbf{E}\mathbf{P}\mathbf{E}] = \sigma^2\mathbf{P}\mathbf{M}_\chi(\mathbf{P})$, of $\mathbf{G}$-norm at most $\mathbb{E}q^2 \le (\mathbb{E}q^4)^{1/2}$.

Decay. $\mathbf{M}_\chi(\mathbf{P}) = \Pi_O\mathbf{P} + \Pi_R\mathbf{P}\Pi_O + \mathrm{tr}(\mathbf{P})\Pi_O + \mathrm{tr}(\Pi_O\mathbf{P})\Pi_R - \Pi_O\,\mathrm{diag}(\mathbf{P})$, with term norms at most $\|\mathbf{P}\|$, $\|\mathbf{P}\|$, $p\|\mathbf{P}\|$, $d_O\|\mathbf{P}\|$, $\|\mathbf{P}\|$; with $\|\mathbf{P}_\lambda\| = 1/(\lambda_{\min}(\mathbf{G}) + \lambda)$ and $\|\boldsymbol{\theta}_\lambda\| \le \|\mathbf{P}_\lambda\|\|\mathbf{c}\|$ for $\Psi = \mathbf{I}$. For $\Psi = \Pi_O$ the O rows of $\mathbf{P}_\lambda$ are $(\mathbf{S}+\lambda\mathbf{I})^{-1}[-\mathbf{C}^\top\mathbf{A}^{-1},\ \mathbf{I}]$, of norm at most $(1+\|\mathbf{A}^{-1}\mathbf{C}\|)/(\lambda_{\min}(\mathbf{S})+\lambda)$, and every term of $\mathbf{M}_\chi(\mathbf{P}_\lambda)\boldsymbol{\theta}_\lambda$ contains a factor $\Pi_O\mathbf{P}_\lambda$ or $\mathbf{P}_\lambda\Pi_O$. □

This is the paper's Theorem on the leading-order bias and its ridge version, with the constants made explicit; it follows the companion book (Chapter 14, Theorems `bias.main` and `bias.ridge`, Proposition `bias.shrink`), whose proofs were checked step by step, including $c_0 = 2.512$ and the fourth moment $64\sigma^4$ of the centred norm. At $p = 11$, $\vartheta = 5$ and $\bar\eta = \frac{1}{2}$ the leading term is at most $0.13\|\boldsymbol{\theta}_\lambda\|_{\mathbf{G}}$ and the remainder at most $0.034(\|\boldsymbol{\theta}_\lambda\|_{\mathbf{G}} + \sigma\sqrt{\mathrm{tr}\,\mathbf{P}})$ (worst direction); in simulation the leading term matches the bias to within 2%. The correction $\tilde{\boldsymbol{\theta}} - \sigma^2\tilde{\mathbf{P}}\mathbf{M}_\chi(\tilde{\mathbf{P}})\tilde{\boldsymbol{\theta}}$ leaves a bias $O(\vartheta^{-4})$ and a fluctuation $O_p(\vartheta^{-3})$ relative to $\|\boldsymbol{\theta}_\lambda\|_{\mathbf{G}}$ (book, Proposition `bias.corr`; checked, with its constant not explicit).

12 S10: the FWS residual variance, binding clipping, and the full-record sensitivity

12.1 The plug-in residual variance of the FWS comparison

The paper's FWS comparison uses $\hat s^2 = \widehat{\mathrm{RSS}}/(n-p)$ with $\widehat{\mathrm{RSS}} = \widehat{yy} - \hat{\mathbf{c}}_2^\top\hat{\mathbf{G}}_2^{-1}\hat{\mathbf{c}}_2$, the case $\lambda = 0$ of Lemma S6. Let ϑ, T, κ_p be as in Section 11 with $\lambda = 0$, and $\hat{\boldsymbol{\beta}} = \mathbf{P}\mathbf{c}$.

Proposition 12.1 (S10a: bias of the plug-in RSS)**.** *If $\bar\eta\vartheta > 1$, then* $\mathbb{E}[\widehat{\mathrm{RSS}} \mid T] - \mathrm{RSS} = -\sigma^2\big(\mathrm{tr}\,\mathbf{P} + \hat{\boldsymbol{\beta}}^\top\mathbf{M}_\chi(\mathbf{P})\hat{\boldsymbol{\beta}}\big) + \rho$ *with*

$$|\rho| \le \frac{1}{\Pr(T)}\Big[\frac{\kappa_p^4}{(1-\bar\eta)\vartheta^4}\big(\|\hat{\boldsymbol{\beta}}\|_{\mathbf{G}}^2 + \sigma^2\,\mathrm{tr}\,\mathbf{P}\big) + \frac{\kappa_p^2}{\vartheta^2}\sigma^2\,\mathrm{tr}\,\mathbf{P} + \frac{2\sqrt{2}\,\kappa_p^2}{\vartheta^2}e^{-p(\bar\eta\vartheta-1)^2/2}\|\hat{\boldsymbol{\beta}}\|_{\mathbf{G}}^2\Big].$$

The leading term is negative: $\hat s^2$ underestimates the residual variance at order σ_2^2.

Proof. With $\boldsymbol{\phi} = \mathbf{G}^{1/2}\hat{\boldsymbol{\beta}}$ and $\mathbf{h} = \mathbf{G}^{-1/2}\mathbf{f}$, $\widehat{\mathrm{RSS}} - \mathrm{RSS} = g - [(\boldsymbol{\phi}+\mathbf{h})^\top(\mathbf{I}+\mathbf{M})^{-1}(\boldsymbol{\phi}+\mathbf{h}) - \|\boldsymbol{\phi}\|^2]$. Expand $(\mathbf{I}+\mathbf{M})^{-1}$ to $\mathbf{M}^4(\mathbf{I}+\mathbf{M})^{-1}$; on the symmetric event T the terms odd in $\mathbf{M}$ vanish, and so do the terms linear in $\mathbf{h}$; $\mathbb{E}\|\boldsymbol{\phi}+\mathbf{h}\|^2 = \|\boldsymbol{\phi}\|^2 + \sigma^2\,\mathrm{tr}\,\mathbf{P}$; $\boldsymbol{\phi}^\top\mathbb{E}[\mathbf{M}^2]\boldsymbol{\phi} = \sigma^2\hat{\boldsymbol{\beta}}^\top\mathbf{M}_\chi(\mathbf{P})\hat{\boldsymbol{\beta}}$. The remainder collects $\boldsymbol{\phi}^\top(\mathbb{E}[\mathbf{M}^2 \mid T] - \mathbb{E}\mathbf{M}^2)\boldsymbol{\phi}$ (bounded as in Theorem 11.1), $\mathbb{E}[\mathbf{h}^\top\mathbf{M}^2\mathbf{h} \mid T] \le \mathbb{E}[q^2]\sigma^2\,\mathrm{tr}\,\mathbf{P}/\Pr(T)$, and the quartic term, at most $\mathbb{E}[q^4\mathbf{1}_T](\|\boldsymbol{\phi}\|^2 + \sigma^2\,\mathrm{tr}\,\mathbf{P})/((1-\bar\eta)\Pr(T))$. $\mathbf{M}_\chi(\mathbf{P}) \succeq 0$ because it is $\sigma^{-2}\mathbb{E}[\mathbf{EPE}]$ with $\mathbf{P} \succ 0$. □

A Monte Carlo check with antithetic pairs ($p = 11$, $\vartheta = 7.0$, 3.5, 1.8) matches the leading term to 0.05–0.65% (`experiments/proofs_check/fws_s2_bias.py`). Rerunning the paper's FWS comparison on the synthetic population (setting I, the same cohorts, releases and FWS random numbers, 200 cohorts per cell) with $\widehat{\mathrm{RSS}}$ corrected by the leading term changes the coverage of the ten slopes by at most 0.0005 in each of the 41 ridge-on cells, and the median half-width by at most 3%: where the ridge is on, the mechanism noise dominates FWS's width, and $\hat s^2$ is floored at 0.1 in up to 49% of releases (`experiments/proofs_check/fws_s2_corrected.py`).

12.2 Binding clipping and the bias-aware interval

When clipping binds, the clipped response is heteroskedastic and $\mathbb{E}[\tilde{\mathbf{y}} \mid \mathbf{X}]$ need not be linear. Take as estimand the projection coefficient $\boldsymbol{\beta}^* = \mathbf{G}^{-1}\mathbf{X}^\top\mathbb{E}[\tilde{\mathbf{y}} \mid \mathbf{X}]$ of the clipped data (fixed design), with the $\tilde y_i$ conditionally independent given $\mathbf{X}$, and let $\bar\sigma^2 \ge \max_i \mathrm{Var}(\tilde y_i \mid \mathbf{X})$.

Proposition 12.2 (S10b: a valid interval under binding clipping)**.** $\mathbb{E}[\boldsymbol{\theta}_\lambda \mid \mathbf{X}] - \boldsymbol{\beta}^* = -\lambda \mathbf{P}_\lambda \Psi \boldsymbol{\beta}^*$, $\mathrm{Cov}(\boldsymbol{\theta}_\lambda \mid \mathbf{X}) \preceq \bar{\sigma}^2 \mathbf{P}_\lambda \mathbf{G} \mathbf{P}_\lambda$, *and the norm bound of Lemma S4 holds for* $\boldsymbol{\beta}^*$. *Consequently Theorem S5(b) holds for* $\boldsymbol{\beta}^*$, *with the same exact and first-order parts, when* $\tilde{s}^2$ *is replaced by* $\bar{\sigma}^2$ *in* $\mathbf{V} = \mathbf{V}_{\mathrm{mech}} + \tilde{s}^2 \tilde{\mathbf{P}} \hat{\mathbf{G}}_2 \tilde{\mathbf{P}}$. *Admissible public choices are:* $\bar{\sigma}^2 = B_y^2$, *with no model; and, when only the response is clipped and* $y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$ *before clipping with* $\mathrm{Var}(\varepsilon_i \mid \mathbf{X}) \le \sigma_0^2$, *any public* $\bar{\sigma}^2 \ge \sigma_0^2$.

Proof. $\mathbb{E}[\mathbf{c} \mid \mathbf{X}] = \mathbf{X}^\top \mathbb{E}[\tilde{\mathbf{y}} \mid \mathbf{X}] = \mathbf{G}\boldsymbol{\beta}^*$, so $\mathbb{E}[\boldsymbol{\theta}_\lambda \mid \mathbf{X}] = \mathbf{P}_\lambda \mathbf{G} \boldsymbol{\beta}^* = \boldsymbol{\beta}^* - \lambda \mathbf{P}_\lambda \Psi \boldsymbol{\beta}^*$; $\mathrm{Cov}(\boldsymbol{\theta}_\lambda \mid \mathbf{X}) = \mathbf{P}_\lambda \mathbf{X}^\top \boldsymbol{\Omega} \mathbf{X} \mathbf{P}_\lambda$ with $\boldsymbol{\Omega} = \mathrm{diag}(\mathrm{Var}(\tilde{y}_i \mid \mathbf{X})) \preceq \bar{\sigma}^2 \mathbf{I}$. Lemma S4 uses no model. The interval is $\tilde{\theta}_j \pm \mathrm{cv}_\alpha(\bar{b}_j / \sqrt{V_{jj}}) \sqrt{V_{jj}}$; if the true standard deviation is $s \le s' = \sqrt{V_{jj}}$ and $|b| \le \bar{b}_j$, then $c = \mathrm{cv}_\alpha(\bar{b}_j / s') s' > \bar{b}_j \ge |b|$ (since $\mathrm{cv}_\alpha(t) > t$ for $\alpha < \frac{1}{2}$), and $s \mapsto \Pr[|sZ + b| \le c] = \Phi((c-b)/s) - \Phi((-c-b)/s)$ has derivative $-[(c-b)\phi((c-b)/s) + (c+b)\phi((c+b)/s)]/s^2 < 0$; so coverage at s is at least coverage at s', which is the guarantee of Theorem S5(b). $|\tilde{y}_i| \le B_y$ gives $\mathrm{Var}(\tilde{y}_i \mid \mathbf{X}) \le B_y^2$. Clipping is 1-Lipschitz, and $\mathrm{Var}\, f(Y) = \frac{1}{2}\mathbb{E}(f(Y) - f(Y'))^2 \le \frac{1}{2}\mathbb{E}(Y - Y')^2 = \mathrm{Var}\, Y$ for 1-Lipschitz f and an independent copy Y', so $\mathrm{Var}(\mathrm{clip}(y_i) \mid \mathbf{X}) \le \sigma_0^2$. □

What remains [open] is a guarantee with the data-driven $\tilde{s}^2$: it estimates an average of the $\mathrm{Var}(\tilde{y}_i \mid \mathbf{X})$ plus a misspecification term, not their maximum, and a sandwich variance would need per-record residuals of O's data, that is, a further release. The public bound B_y^2 is conservative (on the synthetic population $B_y^2 = 8.35$ against a clipped residual variance of about 0.5, which widens the sampling part by about 4.1).

12.3 The full-record sensitivity in closed form

The paper's full-record baseline releases the upper triangle of $\sum_i \mathbf{w}_i \mathbf{w}_i^\top$, $\mathbf{w} = (\dot{\mathbf{x}}, \mathbf{x}_O, y)$, $\dot{\mathbf{x}} = (1, \mathbf{u})$ with an intercept ($\iota = 1$; $\dot{\mathbf{x}} = \mathbf{u}$ and $\iota = 0$ without), every entry with the same noise. Let $R_u = B_R^2 - \iota$, $o = B_O^2$, $w = B_y^2$, and $m_1 = \min(R_u, o)$, $m_2 = \max(R_u, o)$. Define

$$H(a) = \begin{cases} \frac{2}{3}a^2, & |a| \le 3m_1, \\ \frac{1}{2}a^2 + |a| m_1 - \frac{3}{2}m_1^2, & 3m_1 < |a| \le m_1 + 2m_2, \\ 2|a|(m_1 + m_2) - 2(m_1^2 + m_2^2 + m_1 m_2), & |a| > m_1 + 2m_2, \end{cases}$$

$K = \iota(2R_u + 2o) + 2R_u^2 + 2o^2 + 2R_u o$, and, for $t \in [-w, w]$,

$$\Psi_2(t) = K + (\iota + R_u + o)\Big(w + \frac{t^2}{w}\Big) - 2\iota t + \Big(w - \frac{t^2}{w}\Big)^2 + H(\iota + t),$$

and $\Psi_2^{\neg yy}$ the same without the term $(w - t^2/w)^2$.

Theorem 12.3 (S10c: full-record sensitivity)**.** *If* $\mathbf{u}$ *and* $\mathbf{x}_O$ *have dimension at least* 2, *the replace-one* ℓ_2 *sensitivities of the full-record release are* $\Delta^2_{\mathrm{full}} = \max_{|t| \le w} \Psi_2(t)$ *(joint mode),* $\Delta^2_{\mathrm{full},\neg yy} = \max_{|t| \le w} \Psi_2^{\neg yy}(t)$ *(separate mode) and* $\Delta^2_{G,\mathrm{full}} = K + H(\iota)$ *(stage 1).* Ψ_2 *is a piecewise quartic in* t, *so the maximum is attained at* $t = \pm w$, *at a point where* $|\iota + t| \in \{3m_1, m_1 + 2m_2\}$, *or at a real root of the cubic* Ψ_2' *on a piece: a finite enumeration, as for Proposition S1. If a block has dimension* 1, *the same values are valid, conservative upper bounds.*

Proof. The bracket. For a record $\mathbf{w}$ and its replacement $\mathbf{w}'$, write n_k, n_k' for the squared norms of the blocks $k \in \{\mathbf{u}, \mathbf{x}_O, y\}$ and g_k for their inner products. The change of the first moments has squared norm $\iota \sum_k (n_k + n_k' - 2g_k)$; that of the off-diagonal blocks of $\mathbf{w}\mathbf{w}^\top - \mathbf{w}'\mathbf{w}'^\top$ has squared Frobenius norm $\sum_{k<l}(n_k n_l + n_k' n_l' - 2g_k g_l)$; and $\|\mathrm{triu}\, \mathbf{S}_k\|^2 \le \|\mathbf{S}_k\|_F^2 = n_k^2 + n_k'^2 - 2g_k^2$ for each diagonal block. The supremum of the sum (the paper's expression for Δ^2_{full}) is attained: for a block of dimension at

least 2 the two vectors can be chosen with any norms and inner product so that $\mathbf{S}_k$ is diagonal, and for y the bound is exact with $g_y = yy'$.

Norms at the bounds. For fixed g and (y, y') every coefficient of n_k, n'_k ($k = \mathbf{u}, \mathbf{x}_O$) is non-negative, and the admissible set $|g_k| \le \sqrt{n_k n'_k}$ grows with the norms; so the supremum has $n_{\mathbf{u}} = n'_{\mathbf{u}} = R_u$ and $n_O = n'_O = o$. Then, with $s = y^2 + y'^2$ and $t = yy'$, the objective is

$$K + \iota(s - 2t) + (R_u + o)s + (s^2 - 4t^2) + h(g_{\mathbf{u}}, g_O), \qquad h = -2(\iota + t)(g_{\mathbf{u}} + g_O) - 2(g_{\mathbf{u}}^2 + g_O^2 + g_{\mathbf{u}} g_O),$$

over $|g_{\mathbf{u}}| \le R_u$, $|g_O| \le o$ (the term $s^2 - 4t^2 = (y^2 - y'^2)^2$ is absent in separate mode, and every term in y at stage 1).

The inner maximum. h is a concave quadratic; its unconstrained maximizer is $g_{\mathbf{u}} = g_O = -a/3$, $a = \iota + t$, with value $\frac{2}{3}a^2$. If $|a| > 3m_1$, the coordinate with the smaller box is clipped at $-\operatorname{sgn}(a)m_1$ and the other is $-\operatorname{sgn}(a)(|a| - m_1)/2$, feasible while $|a| \le m_1 + 2m_2$, with value $\frac{1}{2}a^2 + |a|m_1 - \frac{3}{2}m_1^2$; beyond, both are clipped, with value $2|a|(m_1 + m_2) - 2(m_1^2 + m_2^2 + m_1 m_2)$. Each candidate satisfies the KKT conditions (for the clipped coordinate the partial derivative $-2a - 4g_1 - 2g_2$ has the sign of $-a$ once $|a| > 3m_1$), so it is the maximum of the concave problem; this is $H(a)$, continuous at the breakpoints.

The outer maximum. For fixed t the objective is strictly increasing in s (its derivative is $\iota + R_u + o + 2s > 0$). Given t, the pairs (y, y') with $yy' = t$ and $y^2, y'^2 \le w$ have $s \le w + t^2/w$, with equality iff one of $|y|, |y'|$ equals B_y; and $(w + t^2/w)^2 - 4t^2 = (w - t^2/w)^2$. So the supremum is $\max_{|t| \le w} \Psi_2(t)$. A configuration with a block of dimension 1 embeds in dimension 2 (collinear vectors), so its supremum is at most the value for dimension 2. □

At unit bounds without an intercept, $\Psi_2(t) = 9 + t^4 + \frac{2}{3}t^2$, so $\Delta^2_{\text{full}} = 32/3$ (at $t = \pm 1$) and $\Delta^2_{G,\text{full}} = 6$, the paper's values. On the simulation population the enumeration gives $\Delta_{\text{full}} = 92.670$ and $\Delta_{G,\text{full}} = 85.816$ (the paper's 92.7 and 85.8). Over 19 bound vectors, with and without an intercept, it agrees with the paper's grid-and-local-search computation (`experiments/baselines/sens_full.py`) to the relative 10^{-12} of that computation's safeguard, in all three modes; with a block of dimension 1 it is above the grid value, as it must be (`experiments/proofs_check/sens_full_closed.py`). Of the 29 benchmark datasets only one (tamielectric, $d = 3$, so $\mathbf{u}$ has dimension 1) uses the dimension-1 case, for which the theorem gives a valid conservative bound and the grid computation remains a numerical refinement.

13 Simulation of Algorithm 6 end to end

`alg6.py` implements Algorithm 6 line by line (stage 1, r_1, $\hat{L}$, λ_I, closed-form λ_O checked against bisection, the κ_Ψ rule, stage 2, gate, ridge solve, bootstrap with $B = 300$, Corr, $\tilde{s}^2$, both intervals, U_{yy} as specified and corrected). `s3_s6_sim.py` runs 2000 replications per scenario: fixed clipped design ($d_R = 4$, $d_O = 6$, $p = 11$, correlated features clipped at the 99% norm quantile), bounded uniform errors ($s = 1$), B_y a true bound so that the working model holds exactly, $\alpha = 0.05$, $\alpha_L = \alpha_U = 0.0025$, $\vartheta^\star = 2.5$, $\kappa_\Psi = 2$, $\sigma_1 = 2.5\sigma_2$, joint mode.

The Wald interval for $\boldsymbol{\theta}_\lambda$ attains its nominal level in every realistic scenario and over-covers slightly in the stress scenario, because the bootstrap variance there is too large. The β interval over-covers whenever the ridge is active: $\bar{b}_j$ is a worst-case bound and $\tilde{s}^2$ is conservative. It is uninformative ($\hat{L} = 0$) without a public bound in two of the three ridge scenarios.

scenario	n	σ_2	med. λ	Ψ	refused	Wald cov. (range over j)	β cov.	$\widetilde{\mathrm{RSS}} - \mathrm{RSS}_\lambda$ / after Corr
off	4000	4	0	–	0	0.946–0.958	0.945–0.968	-0.14 / -0.05 (± 0.20)
ridge **I**	1500	20	477	**I**	0	0.948–0.960	∞ ($\hat{L} = 0$)	-6.83 / -1.68 (± 0.86)
ridge **I**, $B^{\mathrm{pub}} = 1.5\|\boldsymbol{\beta}\|$	1500	20	477	**I**	0	0.948–0.960	1.000	as above
ridge **I**, private $\hat{B}_\beta$	1500	14	167	**I**	0	0.950–0.960	1.000†	-4.49 / -1.31 (± 0.66)
ridge Π_O	3000	10	439	Π_O	0	0.944–0.962	∞ ($\hat{L} = 0$)	-3.03 / -0.07 (± 0.42)
small n, B^{pub}	200	20	1021	**I**	0	0.946–0.962	1.000	-7.43 / -0.41 (± 0.50)
stress, B^{pub}	60	20	319	**I**	0	0.960–0.974	1.000	-23.35 / $+3.05$ (± 0.57)

Table 5: Algorithm 6, 2000 replications each; Wald target $1 - \alpha_{\mathrm{CI}} = 0.955$, β target ≥ 0.95. In every scenario $\Pr[\Omega_L]$, $\Pr[\vartheta_0 \geq \vartheta^\star]$, $\Pr[\hat{L} \leq \lambda_{\min}(\mathbf{G})]$, $\Pr[U_{yy} \geq \mathbb{E}\mathbf{y}^\top\mathbf{y}]$ (both versions) and the agreement of closed-form λ_O with bisection were all 1.000. "Stress" uses $\sigma_1 = 0.01\sigma_2$, so that $\vartheta_0 \approx \vartheta^\star$; it is outside the default design. †The private bound was finite in 96.1% of replications.

14 Corrections to the specification

Each item states what `protocol_v31.tex` says, what is wrong or imprecise, and the corrected statement.

Correction 1 (Sensitivity: $d_O = 1$, the regime-3 algorithm, and the $\mathbf{y}^\top\mathbf{y}$ mode)**.** DeltaRep's comment "for $d_O \geq 2$ these are exact suprema" is correct (Theorem 3.1). For $d_O = 1$ the same numbers are valid but conservative upper bounds; the exact values are in Proposition 3.5, e.g. $\Delta_G^{(1)} = 2\sqrt{ro}$ for $o \leq \frac{11+5\sqrt{5}}{8} r$. In regime 3, replace "one-dimensional maximization, rounded up" by the exact candidate enumeration of Theorem 3.1(iv) (three cubics). Rounding up may be kept as a safeguard. The joint/separate rule of Algorithm 1, line 5, is consistent with Theorem 3.1(ix): $\mathbf{y}^\top\mathbf{y}$ is free iff $\max_t G \leq 4w(r+o)$ in regime 3, always free if $w \leq 2(r+o)$, never free if $w > 3(r+o)$.

Correction 2 (Statements inherited from the earlier paper)**.** (i) "$\mathbf{y}^\top\mathbf{y}$ is free" is false in general: $(1, 1, 5)$ gives $\Delta = 26.07$ against 14.14. (ii) "Using the full Frobenius norm of $\Delta\mathbf{B}$ would be conservative" is false for $d_O \geq 2$: the supremum is the same (Lemma 3.2); the triu maximum is attained only when $\mathbf{D}$ is diagonal. (iii) The worst case is not "$(\mathbf{x}, y) \mapsto (-\mathbf{x}, -y)$" in general; that is regime 2 only. In regime 1 it is $y' = -y$ with $\cos\angle(\mathbf{x}, \mathbf{x}') = -(r-w)/(2o)$ and $\mathbf{D}$ diagonal, and in regime 3 $|y'| < B_y$ when $\mathbf{y}^\top\mathbf{y}$ costs. (iv) "At the optimum the norms are at their bounds" is true for $\|\dot{\mathbf{x}}\|, \|\mathbf{x}\|, \|\mathbf{x}'\|$ (Lemma 3.3), false for $|y'|$ in costly regime 3, and false for $d_O = 1$, where only $\max(|x|, |x'|) = B_O$.

Correction 3 (The radius r_1)**.** The comment "$\Pr[\|\mathbf{G} - \hat{\mathbf{G}}_1\|_{\mathrm{op}} > r_1] \leq \alpha_L$" (Algorithm 6, line 2) is not proved and would need $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} \leq 2\sigma\sqrt{p}$ (Remark 4.2). Replace it by

$$\Pr\big[\lambda_{\max}(\hat{\mathbf{G}}_1 - \mathbf{G}) > r_1\big] \leq \alpha_L,$$

which is proved (Lemma 4.1(c)) and is the only event Algorithm 6 uses. For a two-sided statement use $r_1 = \sigma_1(2\sqrt{p} + 2\sqrt{\ln(2/\alpha_L)})$. Optional improvement: since the law of $\lambda_{\max}(\mathbf{E}_1)/\sigma_1$ depends only on (p, d_A), r_1 may be set to its exact $(1-\alpha_L)$-quantile, computed offline by Monte Carlo with a certified error. For $p = 11$ this is $7.43\sigma_1$ instead of $11.53\sigma_1$, and it costs no privacy.

Correction 4 (λ_O and the Π_O ridge)**.** Replace the bisection by the closed form of Lemma 5.1(c). State explicitly that $\lambda_O < \infty$ iff $\lambda_{\min}(\mathbf{A}) > g^\star + r_1$ (almost surely), and that the Π_O ridge can never certify $\vartheta_0 > \lambda_{\min}(\mathbf{A})/(2\sigma_2\sqrt{p})$.

Correction 5 (U_{yy})**.** As written, U_{yy} bounds $\mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}]$ with probability $\geq 1-2\alpha_U$, not $1-\alpha_U$ (two one-sided events at level α_U each). The Hoeffding term $B_y^2\sqrt{n\ln(1/\alpha_U)/2}$ is itself correct, but it requires the y_i to be conditionally independent given $\mathbf{X}$. Corrected:

$$U_{yy} = \min\Big\{\widehat{yy} + z_{1-\alpha_U/2}\,\sigma_{yy} + B_y^2\sqrt{n\ln(2/\alpha_U)/2},\ nB_y^2\Big\} \qquad \text{or}$$
$$U_{yy} = \min\Big\{\widehat{yy} + \sqrt{2\ln(1/\alpha_U)(nB_y^4/4+\sigma_{yy}^2)},\ nB_y^2\Big\}.$$

Correction 6 (Estimand under clipping)**.** "CI for β_j" must name its target. Either (A) assume the working linear model, conditionally independent and homoskedastic, *for the clipped data*; this holds if the bounds are true bounds, so that clipping never binds. Or (B) target the projection coefficient $\boldsymbol{\beta}^* = \mathbf{G}^{-1}\tilde{\mathbf{X}}^\top\mathbb{E}[\tilde{\mathbf{y}} \mid \tilde{\mathbf{X}}]$, for which the bias bound remains valid but the homoskedastic variance is not, because clipping induces heteroskedasticity, so coverage is not guaranteed. Exactly Gaussian errors cannot hold with $|y| \leq B_y$; normality is asymptotic (Theorem 7.2(vi)).

Correction 7 (First-order accuracy is governed by $\sigma_2\,\mathrm{tr}\,\mathbf{P}_\lambda$, not by $1/\vartheta$ alone)**.** The validity ratio ϑ_0 controls convergence of the Neumann series. The relative size of the neglected mechanism bias in the Wald interval is of order $\sigma_2\,\mathrm{tr}(\mathbf{P}_\lambda) \leq \sqrt{p}/(2\vartheta_0)$, which at $\vartheta^\star = 2.5$, $p = 11$ may be 0.66. Coverage claims are asymptotic as $\vartheta_0 \to \infty$ at fixed p. Report $\sigma_2\,\mathrm{tr}(\tilde{\mathbf{P}})$ as a diagnostic, and consider centring at the bootstrap bias-corrected $2\tilde{\boldsymbol{\theta}} - \overline{\boldsymbol{\theta}^*}$.

Correction 8 (Bootstrap)**.** Without truncation the bootstrap mean and covariance of $\boldsymbol{\theta}^*$ do not exist (Lemma 8.1(e)); the gate $\hat{\vartheta} \geq 1$ does not fix this. Use the symmetric truncation of Lemma 8.1 (keep a draw iff $\|(\hat{\mathbf{G}}_2+\lambda\Psi)^{-1/2}\mathbf{E}^*(\hat{\mathbf{G}}_2+\lambda\Psi)^{-1/2}\|_{\mathrm{op}} \leq \eta$, e.g. $\eta = 1/2$), or the closed forms $b_{\mathrm{RSS}}(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}})$ and $\mathbf{V}_0(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}})$. Use antithetic pairs if Monte Carlo is kept. "$\mathbf{V}_{\mathrm{mech}}$ is the mechanism covariance" holds to first order only (relative $O_p(\vartheta_0^{-1})$).

Correction 9 (The bias bound $\bar{b}_j$)**.** $\bar{b}_j = \lambda\|\Psi\tilde{\mathbf{P}}\mathbf{e}_j\|\hat{B}_\beta$ uses $\tilde{\mathbf{P}}$; the rigorous bound uses $\mathbf{P}_\lambda$. Either state the interval as first order, or inflate: on $\{\|\mathbf{E}_2\|_{\mathrm{op}} \leq r_2\}$ with $r_2 = \sigma_2(2\sqrt{p} + 2\sqrt{\ln(2/\alpha_2)})$ and $\|\tilde{\mathbf{P}}\|r_2 < 1$, $\|\Psi\mathbf{P}_\lambda\mathbf{e}_j\| \leq \|\Psi\tilde{\mathbf{P}}\mathbf{e}_j\| + \|\tilde{\mathbf{P}}\|r_2\|\tilde{\mathbf{P}}\mathbf{e}_j\|/(1-\|\tilde{\mathbf{P}}\|r_2)$, at the price of a further α_2 in the split.

Correction 10 (The private bound on $\|\boldsymbol{\beta}\|$)**.** $\sqrt{U_{yy}/\hat{L}} < \infty$ iff $\lambda_{\min}(\hat{\mathbf{G}}_1) > r_1$. When the ridge is active, $\lambda_{\min}(\hat{\mathbf{G}}_1) < g^\star + r_1$, and in our scenarios $\hat{L} = 0$ in every replication of two of three ridge settings. Even when finite, the bound is of order $\sqrt{(s^2 + \boldsymbol{\beta}^\top\Sigma\boldsymbol{\beta})/\lambda_{\min}(\Sigma)}$, far above $\|\boldsymbol{\beta}\|$. The specification should say that the β interval is informative essentially only with a public B_β^{pub} when the ridge binds.

Correction 11 (Gate and refusals)**.** State whether coverage is unconditional (refusal = no claim) or conditional on passing; the latter loses a factor $1/(1-\Pr[\mathsf{refused}])$. The rigorous refusal bound at the defaults is weak, $\alpha_L + e^{-p(\vartheta^\star-2)^2} = \alpha_L + 0.064$ for $p = 11$, although Monte Carlo gives essentially zero. The notation table should distinguish ϑ_0 (guaranteed $\geq \vartheta^\star$ on Ω_L) from the gate statistic $\hat{\vartheta}$.

Correction 12 (Minor)**.** (i) The reported "shrinkage $-\lambda\tilde{\mathbf{P}}\Psi\tilde{\boldsymbol{\theta}}$" is not $\boldsymbol{\theta}_\lambda - \hat{\boldsymbol{\beta}}_{\mathrm{OLS}}$, which equals $-\lambda\mathbf{G}^{-1}\Psi\boldsymbol{\theta}_\lambda$ exactly; it agrees with it only to first order in $\lambda\|\mathbf{P}_\lambda\Psi\|$, which is not small when the ridge binds. Rename it or report the exact form when $\mathbf{G}$ is well conditioned. (ii) $\tilde{s}^2$ is conservative under the ridge (Section 7); this is a feature, but it should be stated. (iii) The printed $\Delta = 86.18$, $\Delta_G = 77.20$ correspond to unrounded bounds; with $(5.59, 6.24, 2.89)$ the values are 86.15, 77.16. (iv) σ_{yy} is used in Algorithm 6 but not defined: it is σ_2 in joint mode and σ_y in separate mode.

15 Steps not made fully rigorous

(1) $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} \le 2\sigma\sqrt{p}$ (two-sided) is not proved (Remark 4.2); it is not needed by the algorithm.
(2) (Resolved.) The constant of TV_1 is now proved, with an explicit σ^{-3} term (Lemma 2.2(b)).
(3) The boundary $w^*(r, o)$ of the free set in regime 3 is characterized exactly as $\max_t G = 4w(r+o)$, but its interval structure and the table of Remark 3.4 are numerical. Only $2(r + o) \le w^* \le 3(r + o)$ and $w^*(r, 0) = \frac{11+5\sqrt{5}}{8} r$ are proved.
(4) For $d_O = 1$, $\Delta^{(1)}$ (with $\mathbf{y}$) has no closed form; it is an exact two-variable maximization.
(5) Lemma S6: the constants c_k, C_η and the scale $\mathcal{S}_2$ are given in order of magnitude only; the remainder orders are rigorous in form but not in constants.
(6) Proposition S5: replacing the oracle quantities ($\mathbf{V}_0$, $\mathbf{P}_\lambda$ in $\bar{b}_j$, v_j) by plug-ins is first order. Normality of the sampling term is asymptotic. The variation of $\mathbf{V}_0$ with $\boldsymbol{\theta}_\lambda$ is neglected.
(7) The homoskedastic variance is not valid under binding clipping (Correction 6). A public variance bound gives a valid interval (Proposition 12.2); a guarantee with the data-driven $\tilde{s}^2$ is open.
(8) The size of the downward bias of $\lambda_{\min}(\hat{\mathbf{G}}_1)$ is bounded only by the comparison of Lemma 10.2, which is about a factor 1.9 loose on the simulation population; the empirical switch-off factors 1.2 and 1.4 are not bounds (Remark 10.3).

Scripts

All in `v31/proofs/scratch_stats/` (Python 3, NumPy, SciPy; `sim_core.py` imported read-only): `s1_common.py` (exact Δ, pieces, worst-case configuration), `s1_symbolic.py` (exact verification of the polynomial identities of the S1 proof), `s1_check.py` (Table 2), `s1_freeset.py` and `s1_band.py` (free set), `s2_check.py` (Table 3, discrete-Gaussian and TV checks), `s2_goe_ratio.py` (Remark 4.2), `alg6.py` (Algorithm 6), `s3_s6_sim.py` (Tables 4–5); outputs in `*.out`.

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